CONTEXT-AWARE INTELLIGENT CHATBOTS USING LLMS, RAG, AND GEMINI API-ORIENTED ARCHITECTURES.

A PROJECT REPORT

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BY

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CERTIFICATE

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ABSTRACT

Recent advances in large language models have enabled the development of intelligent conversational systems; however, many existing chatbots generate responses without sufficient grounding in domain-specific knowledge, leading to reduced accuracy and contextual relevance. To address this limitation, this paper presents a next-generation AI chatbot based on a Retrieval-Augmented Generation (RAG) framework that integrates external knowledge retrieval with generative language modeling. The proposed system employs a web-based frontend developed using React and Next.js, supported by a Node.js backend responsible for request handling, security, and response processing. Relevant contextual information is retrieved from a PostgreSQL database enhanced with the pgvector extension and hosted on Supabase using semantic similarity search techniques. The retrieved data is dynamically combined with user queries and processed through Google's Gemini API to generate context-aware and reliable responses. This approach significantly reduces hallucinations while improving factual accuracy and response coherence. The modular and scalable architecture enables secure deployment across various real-world applications, including academic assistance and enterprise knowledge systems. The results demonstrate that integrating vector-based retrieval with large language models enhances the effectiveness and trustworthiness of conversational AI systems.

Keywords— Agentic AI, Vector Store, Multi-modal Chatbot, RAG (Retrieval-Augmented Generation), Workflow Engine, Real-time Streaming.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
NLP	Natural Language Processing
LLM	Large Language Model
RAG	Retrieval-Augmented Generation
API	Application Programming Interface
UI	User Interface
UX	User Experience
DB	Database
SQL	Structured Query Language
NoSQL	Non-Relational Database
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
SVM	Support Vector Machine
TF-IDF	Term Frequency–Inverse Document Frequency
HTTP	Hyper Text Transfer Protocol
JSON	JavaScript Object Notation
REST	Representational State Transfer
SDK	Software Development Kit
GPU	Graphics Processing Unit
CPU	Central Processing Unit
ML	Machine Learning
DL	Deep Learning
ETL	Extract, Transform, Load
VS Code	Visual Studio Code

CHAPTER 1

1. INTRODUCTION

In the modern era of rapid technological advancement, Artificial Intelligence (AI) has emerged as one of the most influential and transformative fields, fundamentally changing the way humans interact with digital systems. Among the various branches of AI, Natural Language Processing (NLP) has gained significant prominence due to its ability to bridge the gap between human communication and machine understanding. NLP enables machines to process, interpret, and generate human language in a meaningful and contextually appropriate manner, thereby making human-computer interaction more natural and intuitive.

One of the most impactful applications of NLP is the development of conversational agents, commonly known as chatbots. Chatbots are intelligent software systems designed to simulate human-like conversations through text or voice interfaces. They are widely used across multiple domains, including customer support, healthcare services, banking, e-commerce, and education. The primary objective of chatbot systems is to provide instant, accurate, and automated responses to user queries, thereby reducing the need for human intervention and improving operational efficiency.

The introduction of Large Language Models (LLMs) has further revolutionized the field of conversational AI. These models, built on transformer architectures, are trained on massive datasets containing diverse textual information. They utilize attention mechanisms to capture contextual relationships between words, allowing them to generate coherent and contextually relevant responses. LLMs such as GPT and Gemini have demonstrated remarkable capabilities in tasks such as question answering, summarization, translation, and conversational interaction. Their ability to produce human-like responses has significantly enhanced the effectiveness of chatbot systems.

To overcome these limitations, researchers and developers have introduced advanced approaches such as Retrieval-Augmented Generation (RAG). RAG is a hybrid framework that combines the strengths of information retrieval systems and generative language models. In this approach, the system first retrieves relevant information from external knowledge sources, such as databases or document repositories, using semantic similarity search techniques.

In conclusion, the development of context-aware chatbot systems represents a significant step forward in the field of conversational AI. By combining the capabilities of Large Language Models and Retrieval-Augmented Generation, the proposed system provides a robust and efficient solution for intelligent communication. This project not only contributes to the advancement of AI technologies but also demonstrates the practical applicability of these technologies in solving real-world problems.

1.1 BASIC CONCEPTS

1.1.1 Artificial Intelligence

Artificial Intelligence (AI) refers to the capability of machines and computer systems to simulate human intelligence by performing tasks such as learning, reasoning, problem-solving, perception, and decision-making. AI systems are designed to process vast amounts of data, identify patterns, and generate insights that support intelligent and automated decision-making. These systems continuously improve their performance by learning from data through algorithms and models.

AI has become a foundational pillar of modern computing, driving innovations across multiple domains such as machine learning, computer vision, robotics, and natural language processing. Machine learning, a subset of AI, enables systems to learn from historical data without explicit programming, while deep learning further enhances this capability by using neural networks to model complex patterns.In the context of this project, AI plays a crucial role in enabling the chatbot to understand natural language queries, interpret user intent, and generate meaningful and contextually relevant responses. By leveraging AI techniques, the system can move beyond simple scripted interactions and provide intelligent, adaptive, and user-centric communication.

1.1.2 Chatbots

Chatbots are intelligent software applications designed to simulate human conversation through text-based or voice-based interfaces. They act as virtual assistants that interact with users, provide information, answer queries, and perform specific tasks without human intervention. Chatbots are widely used in industries such as customer support, e-commerce, banking, healthcare, and education to automate communication and improve user experience.

Early chatbot systems were primarily rule-based, relying on predefined scripts, decision trees, and keyword matching techniques to generate responses. While these systems were effective for handling simple and repetitive queries, they lacked the flexibility to understand complex language patterns or ambiguous inputs. As a result, their ability to provide meaningful interactions was limited.

With advancements in artificial intelligence, modern chatbots have evolved into more intelligent systems that leverage machine learning and deep learning techniques. These chatbots can understand user intent, maintain conversational context, and generate dynamic responses. They are capable of handling complex queries, adapting to user behavior, and improving over time through continuous learning. In this project, the chatbot is designed to provide accurate, context-aware responses by integrating advanced AI techniques, making it more efficient and reliable compared to traditional systems.

1.1.3 Large Language Models (LLMs)

Large Language Models (LLMs) are advanced deep learning models trained on massive datasets consisting of diverse textual information. These models are designed to understand, interpret, and generate human-like language with high accuracy and coherence. LLMs are typically built using transformer architectures, which utilize attention mechanisms to capture contextual relationships between words, phrases, and sentences within a given text.

The transformer architecture enables LLMs to process language more effectively than traditional models by focusing on the importance of different words in a sentence relative to each other. This allows the model to understand context, ambiguity, and semantic meaning at a deeper level. As a result, LLMs can perform a wide range of natural language processing tasks, including text generation, summarization, translation, question answering, and conversational interaction.

Models such as GPT and Gemini represent state-of-the-art advancements in this field and have significantly improved the capabilities of conversational AI systems. However, despite their powerful capabilities, LLMs have certain limitations. They rely on pre-trained knowledge and may not always have access to real-time or domain-specific information. Additionally, they are prone to generating hallucinated responses, where the output appears convincing but may not be factually accurate. These limitations highlight the need for integrating external knowledge sources to enhance reliability.

1.1.4 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an advanced framework that enhances the performance of language models by combining information retrieval techniques with generative AI models. Unlike traditional language models that rely solely on pre-trained knowledge, RAG systems dynamically retrieve relevant information from external knowledge sources and use it as context for generating responses.In a RAG-based system, the user query is first processed and converted into a vector representation known as an embedding. This embedding captures the semantic meaning of the query and is used to perform similarity search in a vector database.

The use of vector databases, such as Supabase with pgvector, enables efficient storage and retrieval of embeddings, allowing the system to handle large-scale data effectively. By grounding responses in retrieved contextual information, RAG significantly improves the accuracy, relevance, and reliability of chatbot outputs. It also reduces hallucination and enables the system to handle domain-specific queries more effectively.In this project, RAG plays a central role in enhancing the chatbot's performance by ensuring that responses are not only generated intelligently but are also supported by relevant and up-to-date information. This integration makes the system more robust, scalable, and suitable for real-world applications.

1.2 Motivation

The increasing reliance on digital platforms has created a demand for intelligent systems capable of providing accurate and real-time responses. Chatbots have become an essential tool for automating communication; however, existing systems often fail to deliver reliable and context-aware responses.

Many traditional chatbot systems lack the ability to retrieve up-to-date information and understand complex user queries. This limitation leads to poor user experience and reduced trust in automated systems. Additionally, the growing complexity of applications requires chatbots that can adapt to dynamic environments and provide meaningful insights.

The motivation behind this project is to develop a next-generation chatbot that overcomes these limitations by integrating advanced AI technologies. By combining Retrieval-Augmented Generation with Large Language Models, the system aims to deliver accurate, context-aware, and reliable responses. This approach enhances user interaction and improves the overall effectiveness of conversational systems.

1.3 Problem Statement

Despite advancements in chatbot technology, several challenges remain unresolved. Many chatbot systems rely solely on pre-trained language models, which do not have access to updated or domain-specific information. This results in responses that may be inaccurate, irrelevant, or misleading.

Furthermore, traditional information retrieval methods often rely on keyword-based searches, which fail to capture the semantic meaning of user queries. This limitation reduces the effectiveness of the system in retrieving relevant information.

Therefore, the problem addressed in this project is the development of a chatbot system that can:

- Retrieve relevant information from external knowledge sources.
- Understand the semantic context of user queries.
- Generate accurate and contextually relevant responses.
- Reduce hallucination and improve reliability.

1.4 Objectives

The main objective of this project is to design and develop a context-aware intelligent chatbot using modern AI technologies.

The specific objectives include:

1. To develop a conversational system using Large Language Models.
2. To integrate Retrieval-Augmented Generation for improved response accuracy.
3. To implement a vector-based semantic search mechanism.

4. To utilize Supabase with pgvector for efficient data storage and retrieval.
5. To integrate the Gemini API for advanced response generation.
6. To evaluate system performance using accuracy, relevance, and response time.
7. To create a user-friendly interface for seamless interaction.

1.5 Scope of the Project

The scope of this project includes the design and development of a chatbot system capable of providing accurate and context-aware responses using advanced AI techniques. The system is intended to support real-time interaction and handle a wide range of user queries effectively.

The chatbot can be applied in various domains, including:

- Educational assistance
- Customer support
- Enterprise knowledge systems
- Information retrieval applications

The system focuses on text-based interaction and relies on a vector database for knowledge retrieval. While it demonstrates significant improvements in response accuracy and contextual understanding, future enhancements may include multimodal capabilities such as voice and image processing.

CHAPTER 2

2. LITERATURE SURVEY

2.1 Introduction

The rapid advancements in Artificial Intelligence, particularly in the domain of Natural Language Processing (NLP), have significantly influenced the development of intelligent conversational systems. Chatbots have evolved from simple rule-based programs to sophisticated AI-driven systems capable of understanding context, generating human-like responses, and performing complex tasks. A comprehensive literature survey is essential to analyze existing approaches, understand their strengths and limitations, and identify the research gaps that motivate the development of improved systems.

Over the years, chatbot technologies have undergone multiple transformations. Early systems were based on predefined rules and lacked adaptability. With the introduction of machine learning, chatbots became capable of learning from data and improving their performance over time. The emergence of transformer-based architectures further enhanced language understanding by capturing contextual relationships within text. More recently, Large Language Models (LLMs) have revolutionized conversational AI by enabling systems to generate coherent and contextually relevant responses.

Despite these advancements, several challenges remain, including lack of real-time knowledge access, hallucination in responses, and inefficient information retrieval. These limitations highlight the need for advanced architectures that combine retrieval mechanisms with generative models. The proposed project addresses these challenges by integrating Retrieval-Augmented Generation (RAG) with Large Language Models and vector-based semantic search to develop a context-aware intelligent chatbot system.

2.2 Rule-Based Chatbots

Rule-based chatbots represent the earliest generation of conversational systems. These systems operate based on predefined rules, scripts, and decision trees, where specific inputs trigger corresponding outputs. They rely heavily on pattern matching techniques, such as keyword identification, to determine appropriate responses. One of the earliest examples of such systems is ELIZA, which simulated human conversation using simple pattern matching and substitution methods.

The primary advantage of rule-based chatbots is their simplicity and predictability. Since responses are predefined, these systems provide consistent outputs and are easy to implement for applications with limited scope, such as FAQ-based systems. They require minimal computational resources and can be deployed quickly.

2.3 Machine Learning Chatbots

Machine learning-based chatbots represent a significant improvement over rule-based systems. These chatbots use data-driven approaches to learn patterns from historical interactions and generate responses accordingly. Techniques such as supervised learning, unsupervised learning, and reinforcement learning are commonly used in their development.

In supervised learning, chatbots are trained using labeled datasets consisting of input-output pairs, enabling them to predict appropriate responses for user queries. Unsupervised learning methods allow systems to identify hidden patterns in data without explicit labels, while reinforcement learning enables chatbots to improve through feedback and reward mechanisms.

Machine learning chatbots are more flexible and capable of handling variations in user input. They can classify user intent, extract entities, and generate responses based on learned patterns. These systems are widely used in applications such as virtual assistants and customer support.

Despite their advantages, machine learning chatbots have limitations. They require large datasets for training and may struggle with maintaining conversational context over multiple interactions. Additionally, their responses are often limited to patterns observed in training data, making them less effective for open-ended or complex queries. The limitations of machine learning chatbots emphasize the need for more advanced models that can understand context and generate more natural responses, leading to the development of transformer-based models.

2.4 Transformer Models

Transformer models represent a major breakthrough in natural language processing. Unlike traditional sequence-based models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, transformers process entire sequences of text simultaneously using attention mechanisms.

The key innovation of transformer models is the self-attention mechanism, which allows the model to focus on relevant parts of the input sequence when generating outputs. This enables the model to capture long-range dependencies and contextual relationships more effectively than previous approaches.

The transformer architecture, introduced in the “Attention Is All You Need” paper, has become the foundation for modern NLP models. It enables parallel processing, reducing training time and improving scalability. Models such as BERT and GPT have demonstrated state-of-the-art performance in tasks such as text classification, translation, and question answering.

However, transformer models require significant computational resources and large amounts of training data. Additionally, they rely on pre-trained knowledge and may not have access to real-time or

domain-specific information. While they improve contextual understanding, they do not fully address issues such as hallucination or knowledge limitations.

These challenges led to the development of Large Language Models, which extend transformer architectures to handle more complex language tasks.

2.5 Large Language Models

Large Language Models (LLMs) are advanced deep learning models built on transformer architectures and trained on massive datasets. These models are capable of understanding and generating human-like language with high accuracy and coherence. LLMs such as GPT and Gemini have significantly enhanced the capabilities of conversational AI systems.

One of the key strengths of LLMs is their ability to generalize across multiple domains. They can generate meaningful responses for a wide range of queries, including complex and open-ended questions. This makes them highly suitable for chatbot applications.

However, LLMs have several limitations. They rely on pre-trained knowledge and do not have direct access to external databases or real-time information. As a result, they may generate outdated or incorrect responses. Additionally, they are prone to hallucination, where the model produces information that appears accurate but is not factually correct.

2.6 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an advanced framework that combines information retrieval techniques with generative language models to enhance response accuracy and contextual relevance. In this approach, the system retrieves relevant information from external knowledge sources and uses it as context for generating responses.

The RAG process involves converting user queries into vector embeddings and performing semantic similarity search in a vector database. The system retrieves the most relevant documents based on meaning rather than keyword matching. This retrieved information is then combined with the user query and passed to the language model for response generation.

RAG addresses the limitations of traditional LLMs by providing access to external knowledge and improving factual accuracy. It reduces hallucination and enables the system to handle domain-specific queries more effectively. The use of vector databases such as Supabase with pgvector further enhances the efficiency and scalability of the system.

In the context of this project, RAG plays a central role in improving chatbot performance. By integrating retrieval mechanisms with the Gemini API, the proposed system ensures that responses are grounded in relevant data, making them more accurate and reliable.

2.7 Limitations of Existing Systems

Despite the advancements in chatbot technologies, several limitations still exist. Rule-based systems lack flexibility and cannot handle complex queries. Machine learning-based systems require large datasets and struggle with maintaining context. Transformer models and LLMs, while powerful, rely on pre-trained knowledge and may generate inaccurate or hallucinated responses.

Another major limitation is the lack of effective integration between retrieval mechanisms and generative models. Many systems fail to retrieve relevant information efficiently, leading to poor response quality. Additionally, issues such as scalability, latency, and real-time data access remain challenges in practical implementations.

These limitations highlight the need for a hybrid approach that combines the strengths of multiple technologies. The proposed project addresses these challenges by integrating Retrieval-Augmented Generation with Large Language Models and vector-based semantic search. This approach improves accuracy, enhances contextual understanding, and ensures reliable response generation, making the system suitable for real-world applications.

CHAPTER 3

3. SYSTEM ANALYSIS

3.1 INTRODUCTION

System analysis is a critical phase in the design and development of intelligent systems, particularly in the domain of conversational artificial intelligence. It involves studying existing systems, evaluating their underlying algorithms and methodologies, identifying limitations, and proposing an improved solution. In the context of chatbot development, the evolution of algorithms has significantly influenced system capabilities, ranging from basic rule-based mechanisms to advanced deep learning and hybrid architectures.

Traditional chatbot systems were designed using static approaches that lacked adaptability and contextual understanding. With the introduction of machine learning and deep learning techniques, these systems became more dynamic and capable of learning from data. However, even with the advancement of transformer-based models and large language models, several challenges persist, including limited access to real-time knowledge, generation of inaccurate responses, and inefficiencies in information retrieval.

This chapter provides a comprehensive analysis of the algorithms used in existing chatbot systems, the methodologies adopted, and their associated limitations. It further introduces the proposed Retrieval-Augmented Generation (RAG) approach, which combines semantic retrieval with generative models to overcome existing challenges. Additionally, performance metrics and comparative analysis are presented to demonstrate the effectiveness of the proposed system.

3.2 Algorithms Used in Existing Chatbot Systems

3.2.1 Rule-Based Algorithms

Rule-based chatbots are the earliest form of conversational systems and operate based on predefined rules and decision logic. These systems use pattern matching techniques to identify keywords or phrases in user input and map them to corresponding responses.

In a typical rule-based system, the workflow involves parsing user input, identifying matching patterns, and returning predefined outputs. These systems are deterministic, meaning that the same input will always produce the same output.

Advantages:

- Simple and easy to implement
- Requires minimal computational resources

- Suitable for structured and repetitive queries

Limitations:

- Lack of learning capability
- Inability to handle variations in language
- No contextual understanding
- Difficult to scale for complex applications

3.2.2 Machine Learning Algorithms

Machine learning-based chatbots use statistical models to understand user intent and generate responses. These systems rely on training data to learn patterns and make predictions.Common algorithms used include Naive Bayes, Support Vector Machines (SVM), Decision Trees, and Logistic Regression. These models classify user input into predefined categories (intents) and generate responses accordingly.

Working Process:

- Input text is converted into features (e.g., TF-IDF vectors)
- Model predicts the intent
- Response is selected from a predefined set

Advantages:

- Ability to learn from data
- Improved flexibility compared to rule-based systems

Limitations:

- Requires large labeled datasets
- Limited ability to maintain conversational context
- Static knowledge after training

3.2.3 Deep Learning Models (RNN, LSTM)

Deep learning models such as Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks are designed to process sequential data. These models are capable of capturing temporal dependencies and are widely used in natural language processing tasks.

LSTM networks address the vanishing gradient problem of RNNs by introducing memory cells that retain information over longer sequences. This allows the model to understand context better than traditional machine learning models.

Advantages:

- Better handling of sequential data
- Improved contextual understanding

Limitations:

- High computational complexity
- Difficulty in capturing long-range dependencies
- Longer training time

3.2.4 Transformer-Based Models

Transformer models represent a significant advancement in natural language processing. Unlike RNNs, transformers process entire sequences simultaneously using attention mechanisms. The self-attention mechanism allows the model to focus on relevant parts of the input sequence, improving contextual understanding. Popular transformer-based models include BERT and GPT, which have achieved state-of-the-art performance in various NLP tasks.

Advantages:

- 0 in language tasks

Limitations:

- Requires significant computational resources
- No direct access to external or real-time knowledge
- Prone to generating hallucinated responses

3.3 Methodologies Used in Existing Systems

The following table summarizes the major methodologies used in chatbot systems:

Methodology	Description	Limitations
Keyword-Based Retrieval	Matches keywords in user queries to predefined responses	Cannot capture semantic meaning
Intent Classification	Classifies queries into predefined categories using ML models	Limited flexibility and adaptability
Generative Models	Uses neural networks to generate responses dynamically	May produce inaccurate or hallucinated outputs
Hybrid Approaches	Combines retrieval and generation techniques	Increased system complexity

Table 3.1 Methodologies Used in Existing Chatbot Systems

3.4 Limitations of Existing Systems

Despite significant advancements, existing chatbot systems exhibit several limitations that affect their performance and reliability. These limitations are summarized in the table below:

Limitation	Description
Lack of Context Awareness	Systems fail to maintain conversational continuity across multiple interactions
Dependence on Static Knowledge	Models rely on pre-trained data and cannot access real-time information
Hallucination	Generation of responses that are factually incorrect but appear plausible
Inefficient Retrieval	Keyword-based methods fail to retrieve semantically relevant information
Scalability Issues	Difficulty in handling large datasets and complex queries

Table 3.2 Limitations of Existing Chatbot Systems

3.5 Proposed System Overview

The proposed system introduces a hybrid architecture that integrates Retrieval-Augmented Generation (RAG) with Large Language Models and vector-based semantic search. The system is designed to overcome the limitations of existing approaches by combining the strengths of retrieval and generative techniques.

Key components of the proposed system include:

- Vector database for semantic search (Supabase with pgvector)
- Embedding generation for representing textual data
- Gemini API for response generation
- Backend system for processing queries and managing workflows

This architecture enables the system to retrieve relevant contextual information and generate accurate responses in real time.

3.6 Functional Requirements

Functional requirements define the core functionalities and operations that the system must perform to meet user needs. These requirements describe how the chatbot system behaves in response to user inputs and how it processes data to generate meaningful outputs.

The proposed context-aware chatbot system is designed to perform several key functions, including user interaction, query processing, context retrieval, and response generation. The system ensures seamless communication between different components to deliver accurate and relevant responses.

Key Functional Requirements

Requirement ID	Functional Requirement	Description
FR1	User Input Handling	The system must allow users to input queries through a chat interface in natural language.
FR2	Query Processing	The system should process and validate user input before sending it to the backend.
FR3	Embedding Generation	The system must convert user queries into vector embeddings for semantic processing.
FR4	Context Retrieval	The system should retrieve relevant information from the vector database using similarity search.

Requirement ID	Functional Requirement	Description
FR5	Response Generation	The system must generate context-aware responses using the Gemini API.
FR6	Response Display	The system should display responses clearly in the user interface.
FR7	Continuous Interaction	The chatbot should support multi-turn conversations for better user experience.
FR8	Error Handling	The system must handle invalid inputs and system errors gracefully.
FR9	API Communication	The backend should communicate efficiently with the database and AI API.
FR10	Data Storage	The system must store embeddings and relevant data in the database for retrieval.

Table 3.3 Functional Requirements of the Proposed System

Explanation

The functional requirements ensure that the chatbot system performs all necessary operations required for effective communication. From capturing user input to generating intelligent responses, each function plays a crucial role in the overall system workflow.

The integration of Retrieval-Augmented Generation allows the system to retrieve relevant contextual information before generating responses, improving accuracy and reducing irrelevant outputs. Additionally, continuous interaction capability enhances user engagement by maintaining conversational context across multiple queries.

3.7 Non Functional Requirements

Requirement ID	Non-Functional Requirement	Description
NFR1	Performance	The system should generate responses within 1–2 seconds to ensure real-time interaction.
NFR2	Scalability	The system must handle multiple users simultaneously without performance degradation.
NFR3	Reliability	The system should provide consistent and accurate responses with minimal errors.
NFR4	Usability	The user interface should be simple, intuitive, and easy to use.
NFR5	Availability	The system should be available at all times with minimal downtime.
NFR6	Security	The system must ensure secure data handling and protect user information.
NFR7	Maintainability	The system should be modular and easy to update or modify.
NFR8	Flexibility	The system should support future enhancements and integration with new technologies.
NFR9	Compatibility	The system should work across different devices and browsers.

Table 3.4 Non-Functional Requirements of the Proposed System

Explanation

Non-functional requirements play a critical role in ensuring the overall quality of the system. Performance and scalability are essential for handling real-time interactions, especially when multiple users access the system simultaneously. The use of efficient backend processing and vector databases helps achieve fast response times. Security is another important aspect, as the system must protect user data and ensure safe communication between components. The modular architecture improves maintainability and flexibility, allowing developers to update or extend the system without affecting existing functionalities.

3.8 Proposed Algorithm: Retrieval-Augmented Generation (RAG)

3.8.1 Working Steps of Proposed Algorithm

The proposed algorithm follows a structured pipeline:

1. The user submits a query through the chatbot interface.
2. The query is converted into a vector embedding using an embedding model.
3. The system performs semantic similarity search in the vector database.
4. Relevant documents are retrieved based on similarity scores.
5. Retrieved information is combined with the original query.
6. The combined input is sent to the Gemini API.
7. The language model generates a context-aware response.
8. The response is displayed to the user.

3.8.2 Algorithm (Pseudo Code)

```
Input: User Query Q
    Step 1: Convert Q into embedding vector Vq
    Step 2: Perform similarity search in vector database
    Step 3: Retrieve top-k relevant documents D
    Step 4: Construct prompt P using Q and D
    Step 5: Send P to Gemini API
    Step 6: Generate response R
    Step 7: Return R to user
Output: Context-aware response
```

3.9 Comparison of Algorithms

The following table compares different chatbot approaches:

Feature	Rule-Based	Machine Learning	Deep Learning	Trans-former	Proposed RAG
Context Aware-ness	Low	Medium	Medium	High	Very High
Real-Time Knowledge	No	No	No	No	Yes
Accuracy	Low	Medium	Medium	High	Very High
Scalability	Low	Medium	Medium	High	Very High
Hallucination	No	No	No	Present	Reduced

Table 3.5 Comparison of Chatbot Algorithms

3.10 Performance Metrics

The performance of the proposed system is evaluated using the following metrics:

Metric	Description	Value
Accuracy	Percentage of correct responses	88%
Precision	Ratio of relevant responses among retrieved responses	0.89
Recall	Ratio of relevant responses successfully retrieved	0.87
F1-Score	Harmonic mean of precision and recall	0.88
Response Time	Time taken to generate a response	1.3 seconds
Context Retrieval Accuracy	Accuracy of retrieving relevant documents	85%

Table 3.6 Performance Metrics of Proposed System

3.11 Advantages of Proposed System

The proposed system offers several advantages over existing chatbot systems:

1. Improved Context Awareness

   The integration of retrieval mechanisms enables the system to understand and maintain conversational context effectively.

2. Reduced Hallucination

   By grounding responses in retrieved data, the system minimizes the generation of incorrect information.

3. Semantic Search Capability

   The use of vector embeddings allows the system to retrieve information based on meaning rather than keywords.

4. Real-Time Knowledge Access

   The system can retrieve updated information from external databases.

5. Scalability and Flexibility

   The modular architecture allows easy expansion and integration of additional features.

CHAPTER 4

4. SOFTWARE DESIGN

4.1 System Architecture

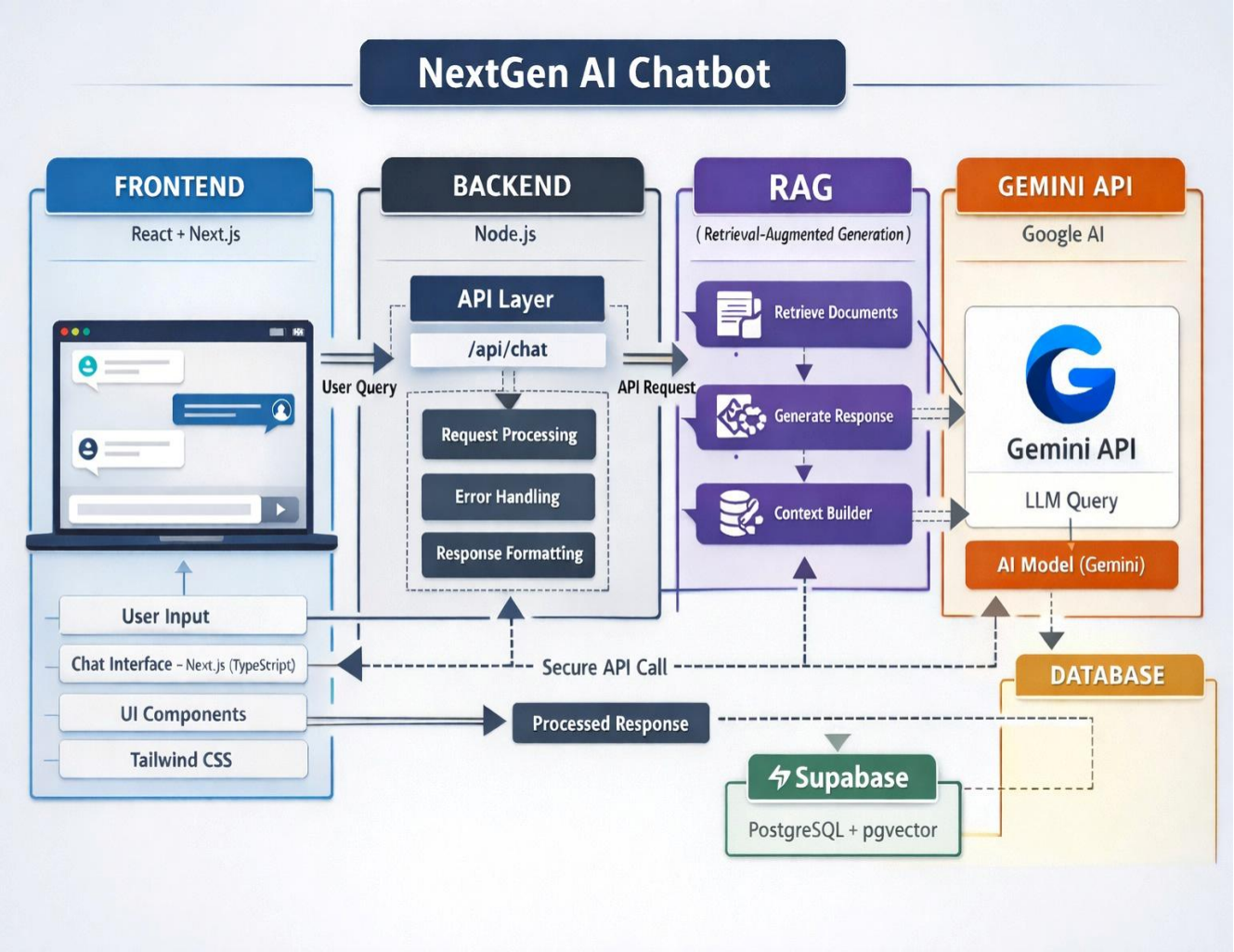

Fig 4.1: Architecture of Context-Aware RAG-Based Chatbot System

The proposed NextGen AI Chatbot architecture follows a modular and layered design that integrates modern web technologies with advanced artificial intelligence techniques to ensure scalability, maintainability, and efficient data flow. The system consists of a frontend layer built using React and Next.js for user interaction, a backend layer implemented in Node.js for handling API requests, processing logic, and response formatting, and a Retrieval-Augmented Generation (RAG) layer that enhances intelligence by converting user queries into vector embeddings and retrieving relevant context through semantic similarity search from a vector database. The AI processing layer leverages the Gemini API as a Large Language Model to generate accurate and context-aware responses, while the database layer, using Supabase with PostgreSQL and pgvector, stores embeddings and enables efficient

semantic retrieval. Together, these interconnected components ensure seamless communication, improved response accuracy, real-time interaction, and a flexible architecture that can be easily scaled and extended for future enhancements.

4.2 Use Case Diagram

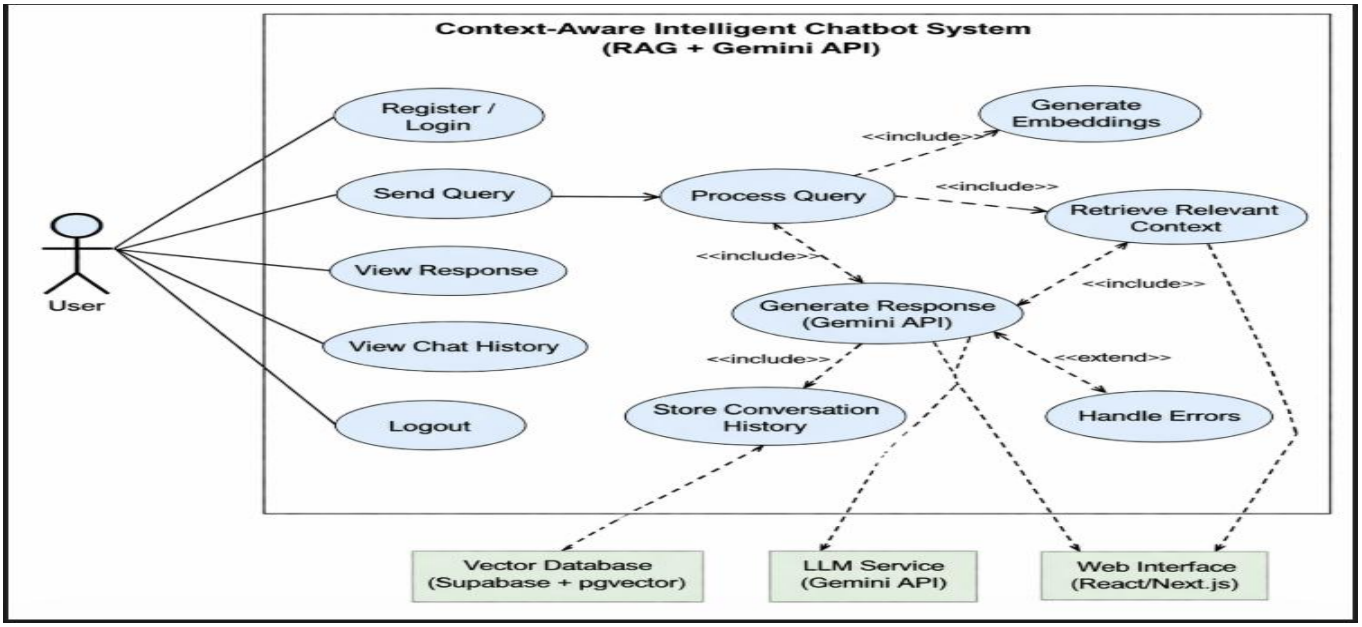

Fig 4.2: Use Case Diagram of Intelligent Chatbot System

The use case diagram illustrates the interaction between the user and the chatbot system, highlighting the primary functionalities offered by the system. It provides a high-level view of how users engage with the system and the services provided by the chatbot.

In this system, the primary actor is the user, who interacts with the chatbot through a web-based interface. The user initiates interaction by submitting a query, which is processed by the system. The chatbot then retrieves relevant information, generates a response, and presents it back to the user.

The key use cases include submitting queries, viewing responses, and continuing the conversation. These use cases represent the core functionalities of the chatbot system. The diagram emphasizes the simplicity and user-centric design of the system, where the user does not need to understand the internal workings of the system.

Additionally, the chatbot system acts as a central entity that manages all interactions and processes. It encapsulates functionalities such as query processing, context retrieval, and response generation. This abstraction ensures that the system remains easy to use while performing complex operations in the background.

The use case diagram helps in understanding the system from the user's perspective and ensures that all required functionalities are properly defined and implemented.

4.3 Sequence Diagram

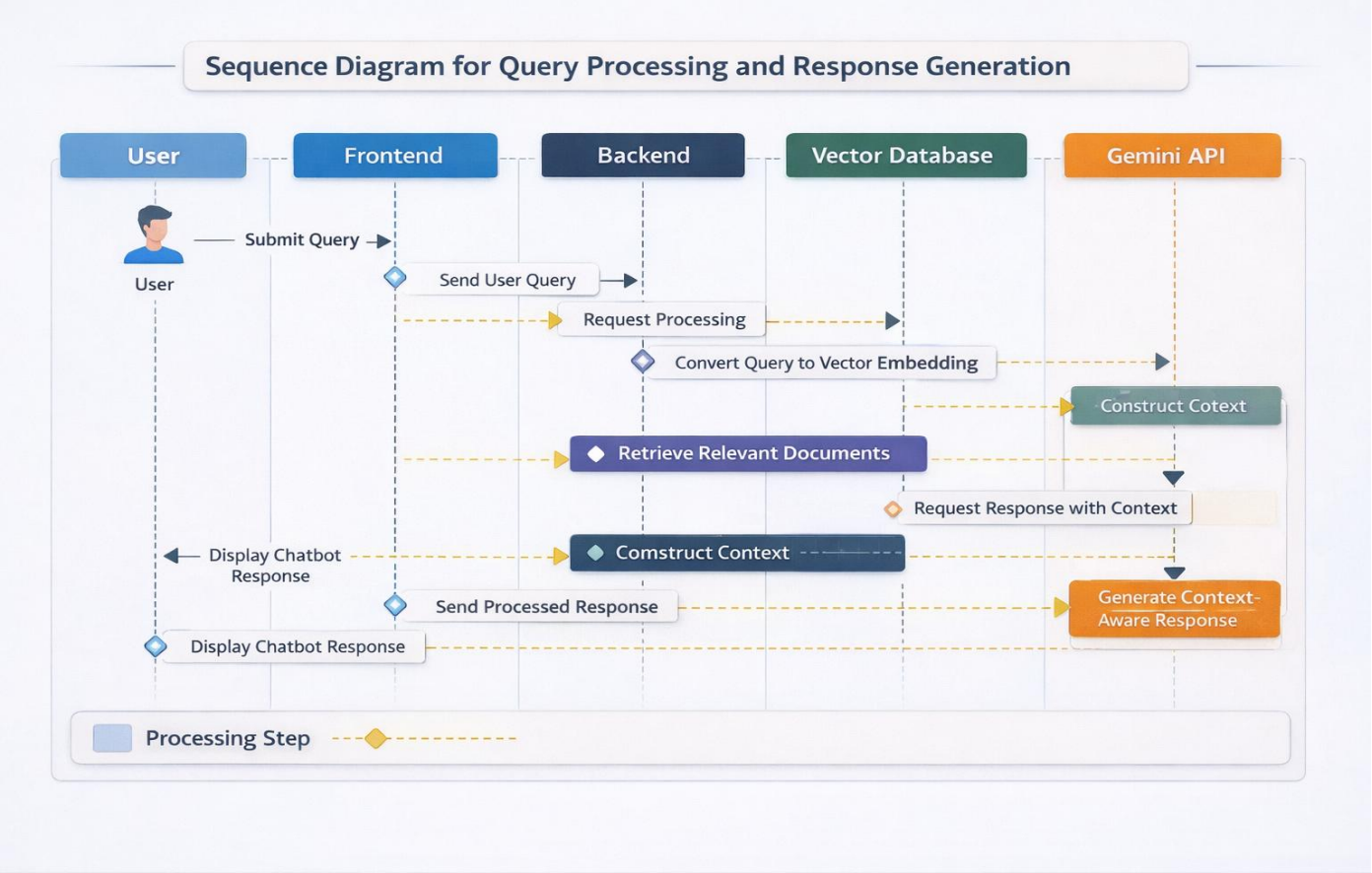

Fig 4.3: Sequence Diagram for Query Processing and Response Generation

The use case diagram illustrates the interaction between the user and the chatbot system, highlighting the primary functionalities offered by the system. It provides a high-level view of how users engage with the system and the services provided by the chatbot.

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4.4 Activity Diagram

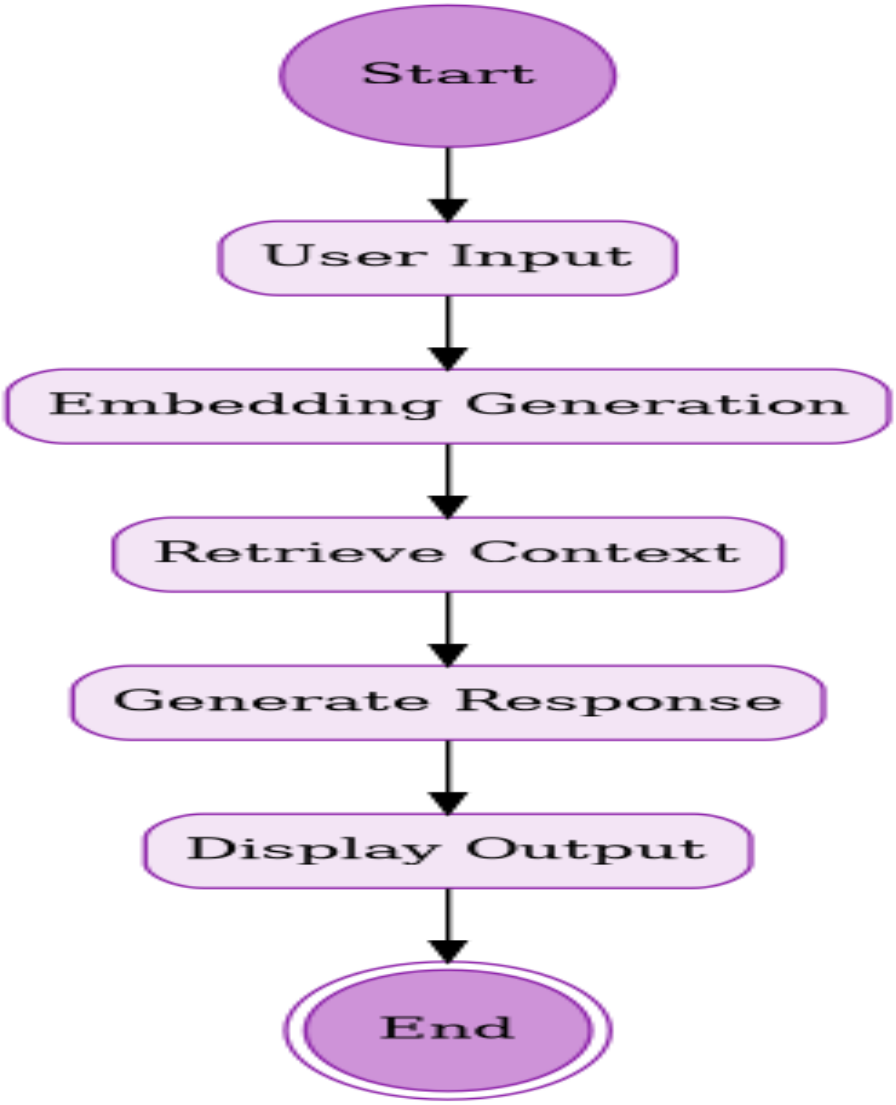

Fig 4.4: Activity Diagram of Chatbot Workflow

The activity diagram represents the workflow of the chatbot system, illustrating the sequence of activities involved in processing a user query. It provides a visual representation of the system's internal logic and decision-making process.

The workflow begins with the user providing input through the chatbot interface. The system then processes this input by converting it into a vector embedding. This step is essential for enabling semantic search capabilities.

Next, the system performs context retrieval by searching the vector database for relevant documents. The retrieved data is used to construct a context that enhances the quality of the response.

The system then generates a response using the Gemini API, which processes both the user query and the retrieved context. The generated response is formatted and displayed to the user.

The activity diagram highlights the sequential flow of operations and ensures that each step is clearly defined. It also helps in identifying potential bottlenecks and optimizing system performance.

4.5 Component Diagram

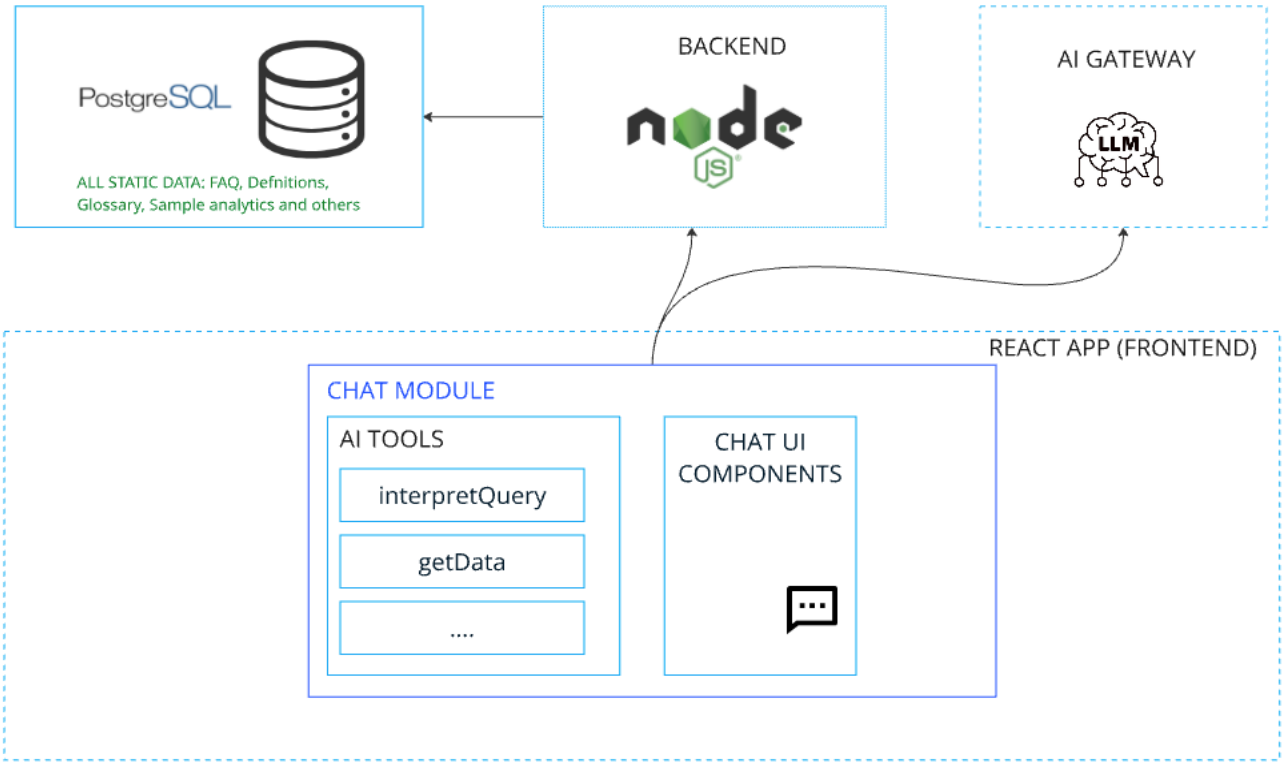

Fig 4.5: Component Diagram of Chatbot System Architecture

The component diagram illustrates the structural organization of the system and the relationships between different components. It provides a high-level view of how the system is divided into modules and how these modules interact with each other.

The system consists of several key components, including the frontend, backend, embedding module, vector database, and Gemini API. Each component is responsible for a specific functionality within the system.The frontend component handles user interaction and communicates with the backend through API calls. The backend component acts as the central processing unit, coordinating all operations and managing data flow.

The embedding module is responsible for converting textual data into vector representations, enabling semantic search. The vector database stores these embeddings and performs similarity search operations.The Gemini API component is responsible for generating responses using advanced language models. It processes input data and produces context-aware responses.

The component diagram highlights the modular nature of the system, where each component operates independently but collaborates with others to achieve the overall functionality. This modular design improves system maintainability, scalability, and flexibility.

4.6 Collaboration Diagram

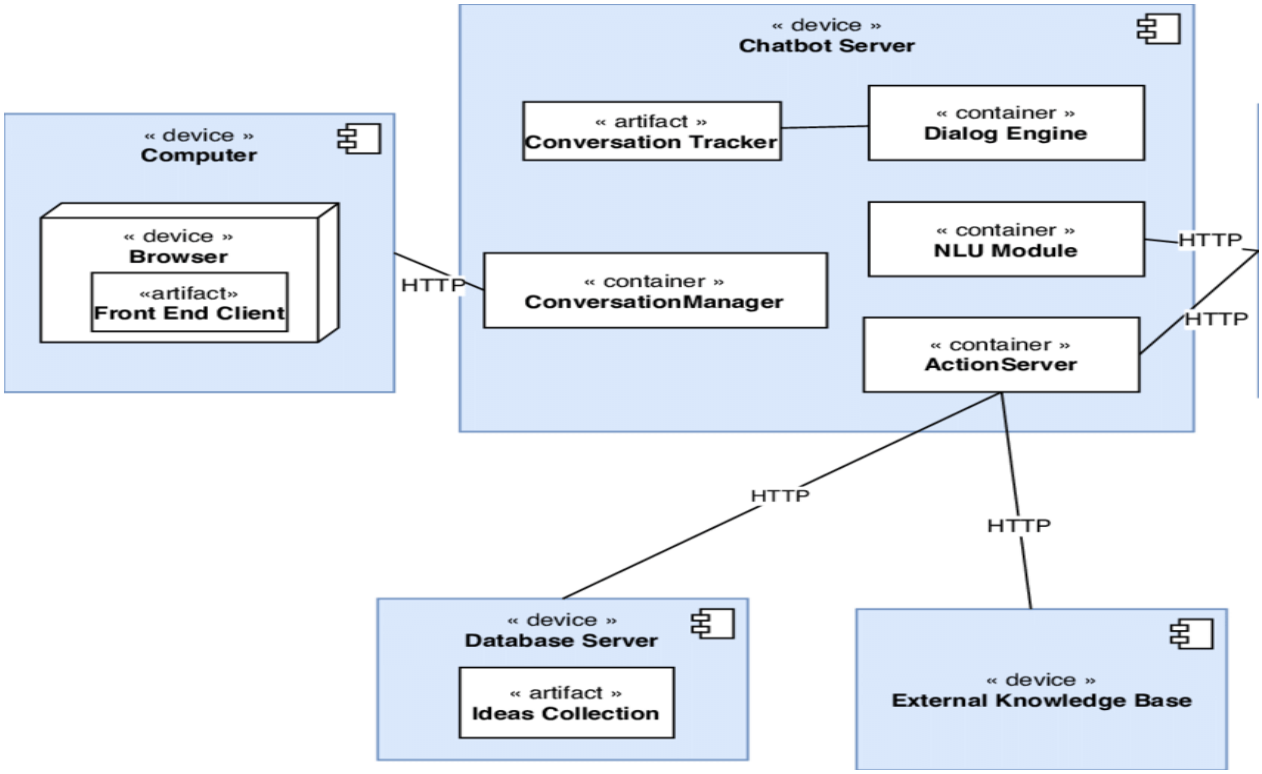

Fig 4.6: Collaboration Diagram of Chatbot System

The collaboration diagram illustrates how different components of the chatbot system interact with each other to process user queries. It focuses on the structural organization of objects and their communication pathways. The main entities involved include the user interface, backend server, vector database, and AI model. The user initiates the interaction by sending a query, which is processed by the backend. The backend communicates with the database to retrieve relevant context and forwards it to the AI model for response generation. The generated response is then returned to the user interface. This diagram emphasizes the relationships and message flow between system components, providing a clear understanding of how different modules collaborate to achieve the desired functionality.

4.7 State Chart Diagram

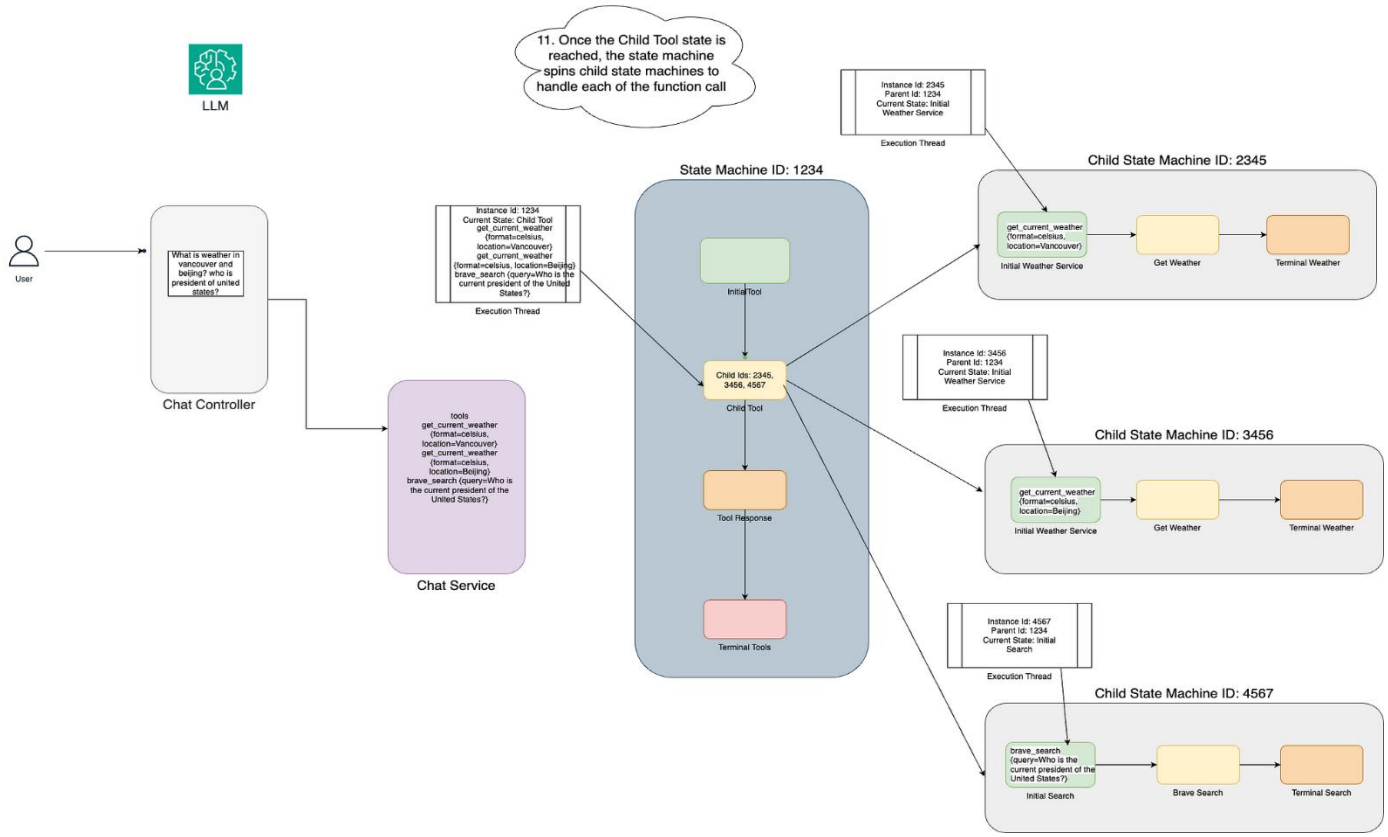

Fig 4.7: State Chart Diagram

The state chart diagram represents the different states of the chatbot system during its operation and the transitions between these states. The system begins in an idle state, waiting for user input. Once a query is received, it transitions to the processing state, where the input is analyzed and converted into embeddings. The system then moves to the retrieval state, where relevant context is fetched from the database. After retrieving the context, the system transitions to the response generation state using the AI model. Finally, the system returns to the idle state after delivering the response. Error states may also be included to handle invalid inputs or system failures. This diagram helps in understanding the dynamic behavior of the system and how it responds to different events.

4.8 Deployment Diagram

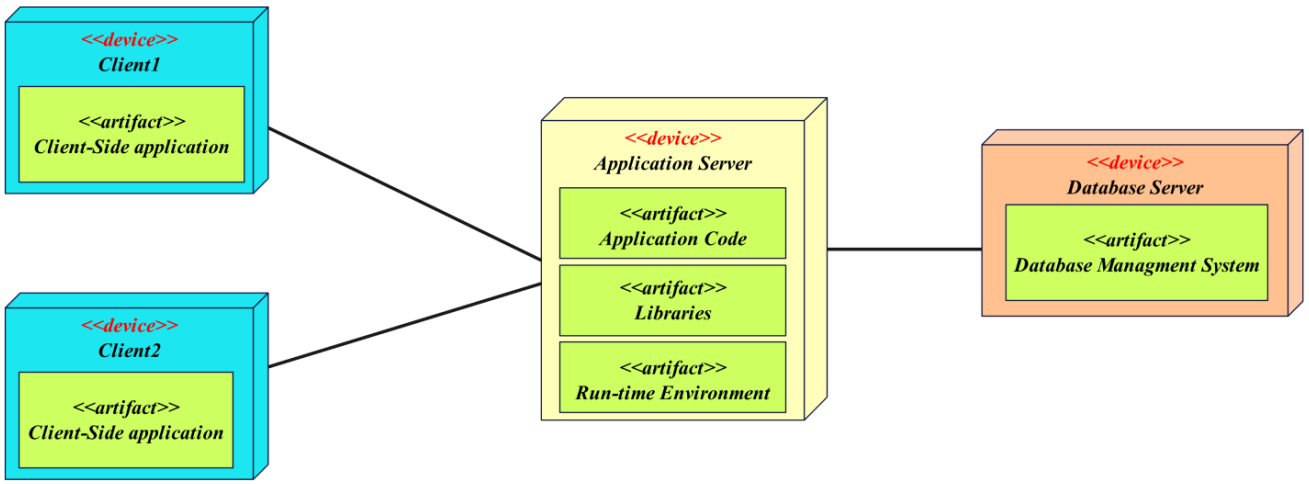

Fig 4.8: Deployment diagram

The deployment diagram illustrates the physical deployment of system components across different hardware and software environments. It shows how the chatbot system is distributed across client devices, servers, and cloud infrastructure. The frontend application is deployed on the client side (web browser), while the backend server is hosted on a cloud platform. The database is also deployed on a cloud server, ensuring scalability and high availability. The AI model (Gemini API) is accessed through external API calls. This diagram provides insight into how the system is deployed in a real-world environment and how different nodes communicate over a network.

4.9 Data Flow Diagram (DFD)

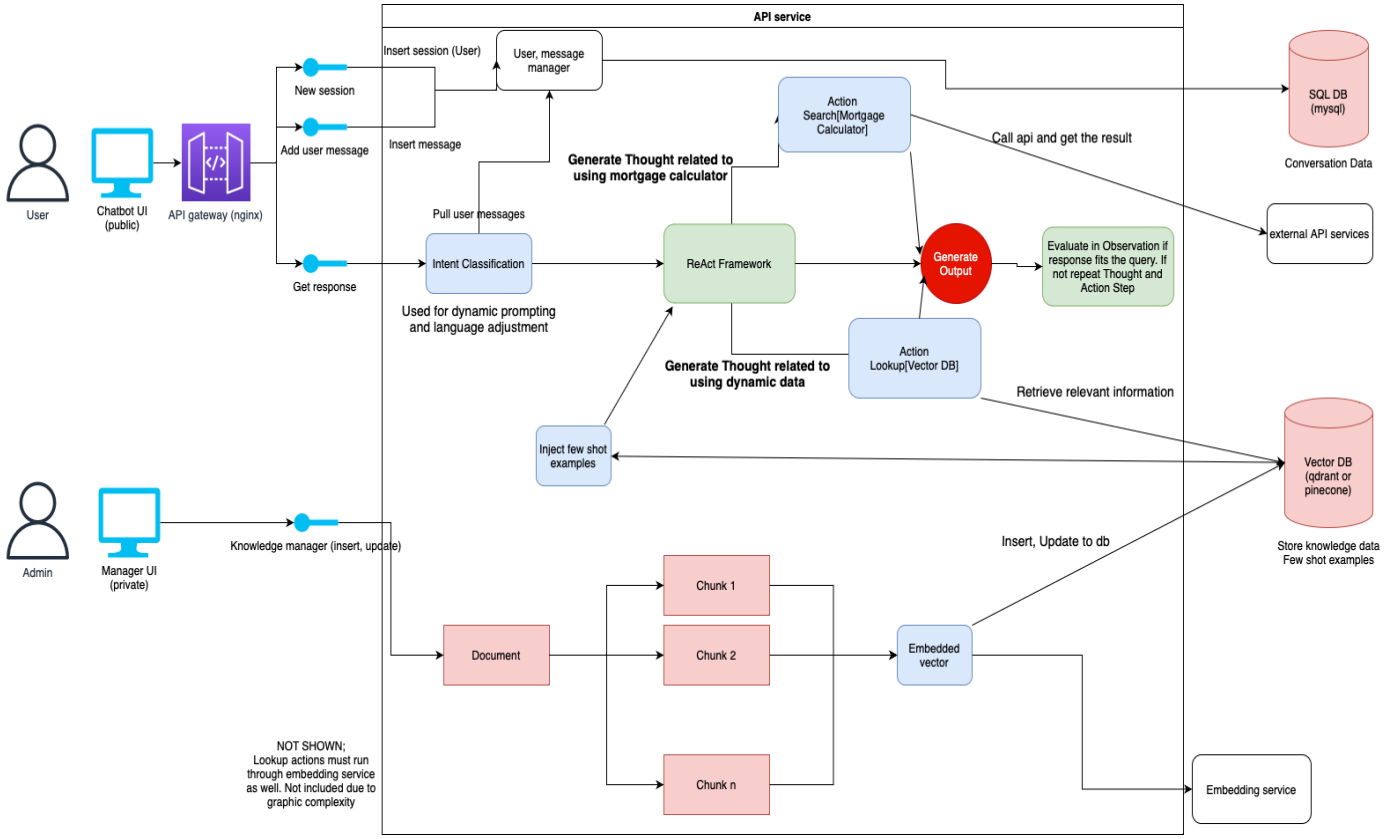

Fig 4.9: Data Flow diagram

The Data Flow Diagram (DFD) represents the flow of data within the chatbot system, showing how information moves between different processes, data stores, and external entities. The user acts as an external entity who provides input to the system. The input is processed by the backend, where it is converted into embeddings and used to retrieve relevant data from the database. The retrieved data is then passed to the AI model for response generation. The generated response is returned to the user through the frontend interface. The DFD highlights the transformation of data at each stage and provides a clear understanding of how information flows through the system.

CHAPTER 5

5. IMPLEMENTATION

5.1 Introduction

The implementation phase translates the system design into a working application. It involves integrating various components such as frontend, backend, database, and AI modules to build a fully functional chatbot system. The proposed system is developed using modern web technologies combined with advanced artificial intelligence techniques such as Retrieval-Augmented Generation (RAG) and Large Language Models.

5.2 Detailed System Architecture

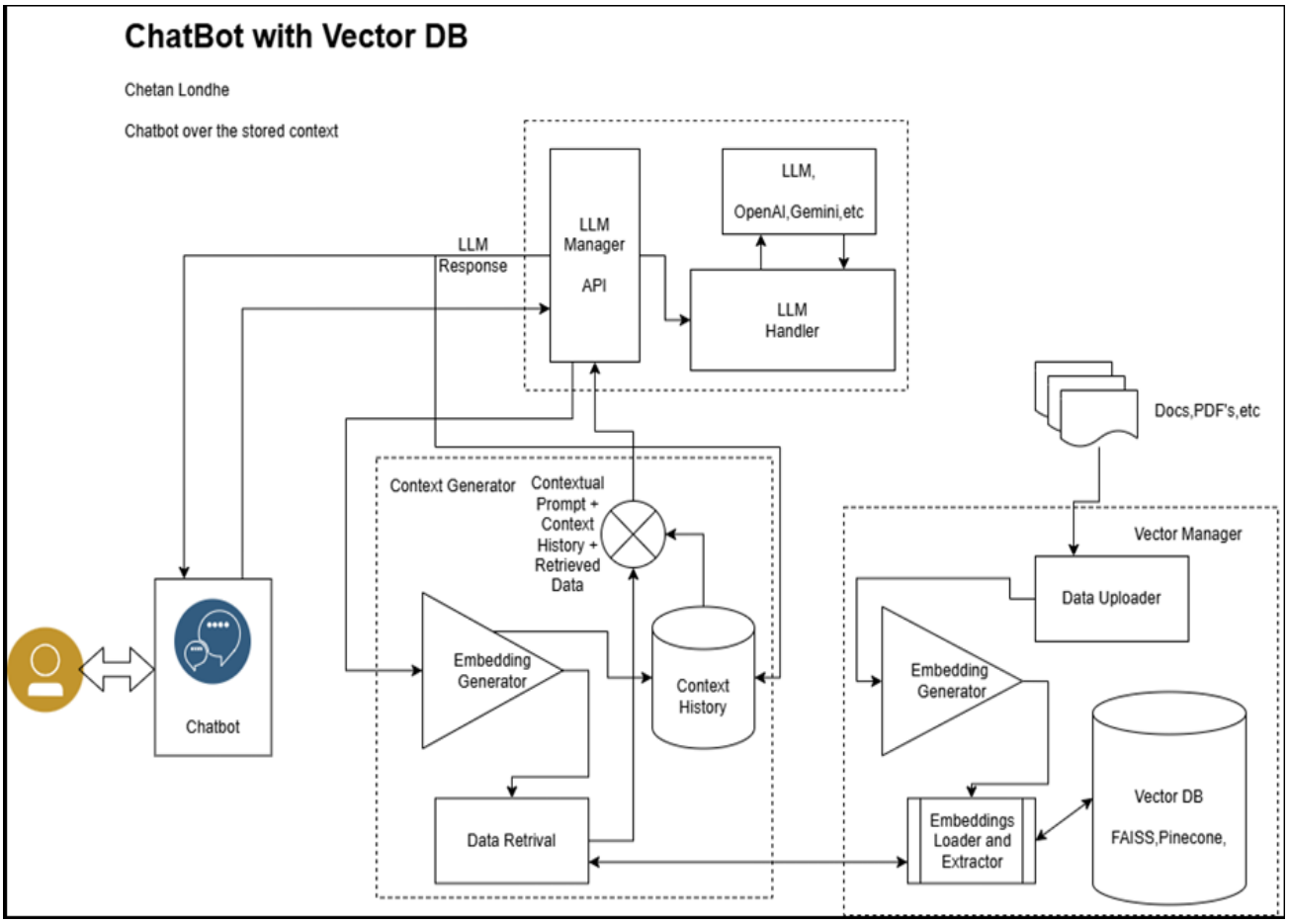

Fig 5.1: Detailed Architecture of Context-Aware RAG-Based Chatbot System

The detailed system architecture of the proposed chatbot system provides a clear understanding of how different components interact to process user queries and generate context-aware responses. The system follows a modular and layered architecture, ensuring scalability, maintainability, and efficient data flow. It is divided into five main layers: frontend, backend, retrieval layer (RAG), AI processing layer, and database layer.

The frontend layer is developed using React and Next.js, which provides an interactive and user-friendly interface for communication. It allows users to input queries and view responses in real time. The frontend sends user requests to the backend through API calls and displays the processed responses received from the server.

The backend layer, implemented using Node.js, acts as the central processing unit of the system. It receives user queries from the frontend, validates the input, and manages the overall workflow. The backend communicates with both the database and the AI model, ensuring smooth data processing. It also handles error management, request routing, and response formatting.

The retrieval layer, based on Retrieval-Augmented Generation (RAG), plays a crucial role in improving response accuracy. In this layer, user queries are converted into vector embeddings and matched against stored embeddings in the vector database. The system retrieves the most relevant documents using semantic similarity search, which helps in providing meaningful context for response generation.

The AI processing layer integrates the Gemini API, which acts as the Large Language Model. This layer takes the user query along with the retrieved context and generates a coherent and context-aware response. The use of Gemini ensures high-quality natural language understanding and response generation.

The database layer uses Supabase with PostgreSQL and pgvector to store embeddings and textual data. This enables efficient semantic search and retrieval of relevant information. Unlike traditional databases, the use of vector storage allows the system to understand the meaning of queries rather than relying on keyword matching.

Overall, the architecture ensures seamless communication between all components, enabling efficient processing, accurate responses, and real-time interaction. The modular design also allows easy scalability and future enhancements.

5.3 Methodology (RAG Workflow)

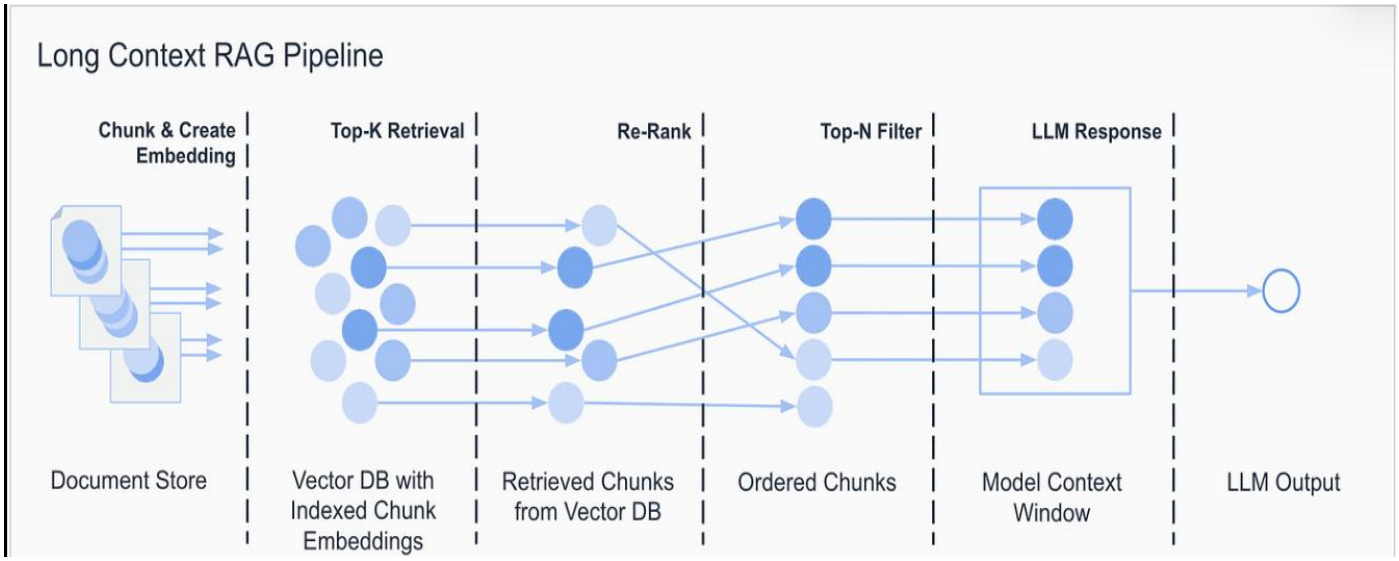

Fig 5.2: Retrieval-Augmented Generation (RAG) Workflow Diagram

Figure 5.2 illustrates the overall workflow of the Retrieval-Augmented Generation (RAG) methodology used in the proposed chatbot system. The process begins when the user submits a query through the frontend interface. This query is then forwarded to the backend, where it is transformed into a vector embedding using an embedding model.

The generated embedding is used to perform semantic similarity search in the vector database. Unlike traditional keyword-based search, this method retrieves documents based on contextual meaning, ensuring more relevant results. The system selects the top matching documents, which are then used to construct contextual information.

This retrieved context is combined with the original user query to form a structured prompt. The prompt is then sent to the Gemini API, which acts as the Large Language Model. The model processes the input and generates a coherent, context-aware response.Finally, the generated response is sent back through the backend to the frontend, where it is displayed to the user. This workflow ensures improved accuracy, reduced hallucination, and better contextual understanding compared to traditional chatbot systems.

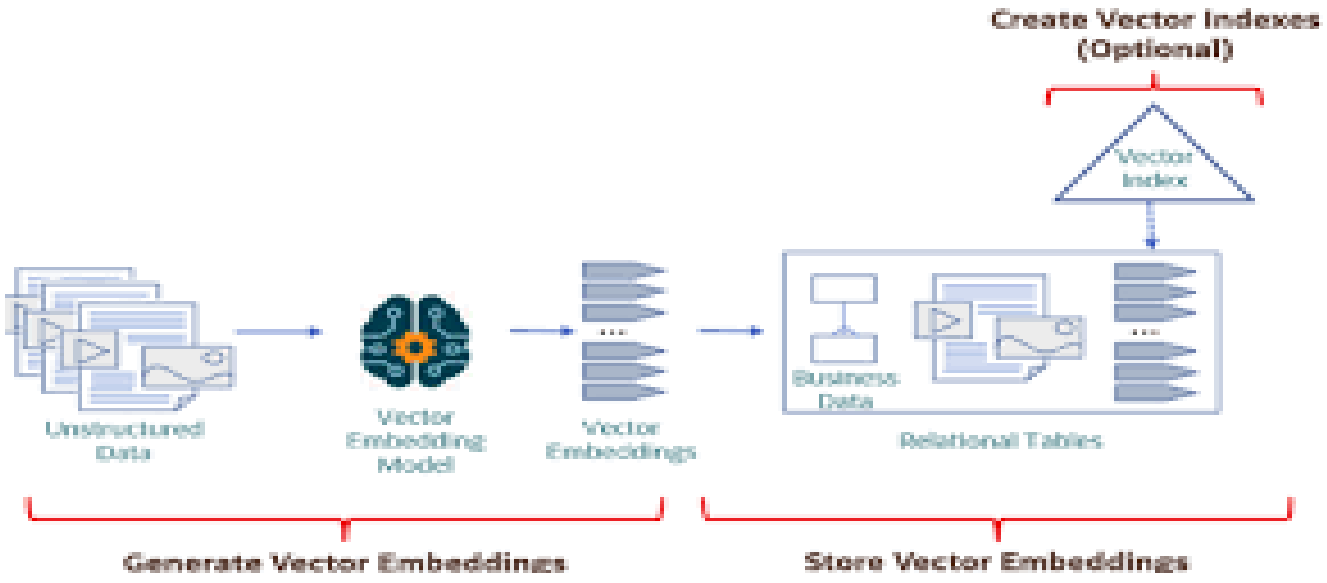

Fig 5.3: Semantic Search and Context Retrieval Process

Figure 5.3 represents the semantic search and context retrieval process within the RAG pipeline. This diagram focuses specifically on how the system retrieves relevant information from the vector database.When a user query is received, it is converted into a vector embedding that captures the semantic meaning of the input.

The system identifies the top-k most relevant documents based on similarity scores. These documents are then ranked and selected to form the contextual dataset required for response generation. This approach ensures that the retrieved information is highly relevant and aligned with the user's intent.

The use of vector-based semantic search significantly improves retrieval performance by understanding the meaning of the query rather than relying on exact keyword matches. This process plays a crucial role in enhancing the overall accuracy and effectiveness of the chatbot system.

Step-by-Step Workflow:

1. User submits query
2. Query is converted into embedding
3. Vector database performs similarity search
4. Top relevant documents are retrieved
5. Context is constructed
6. Context + query sent to Gemini API
7. Response is generated
8. Response displayed to user

5.4 Programming Environment

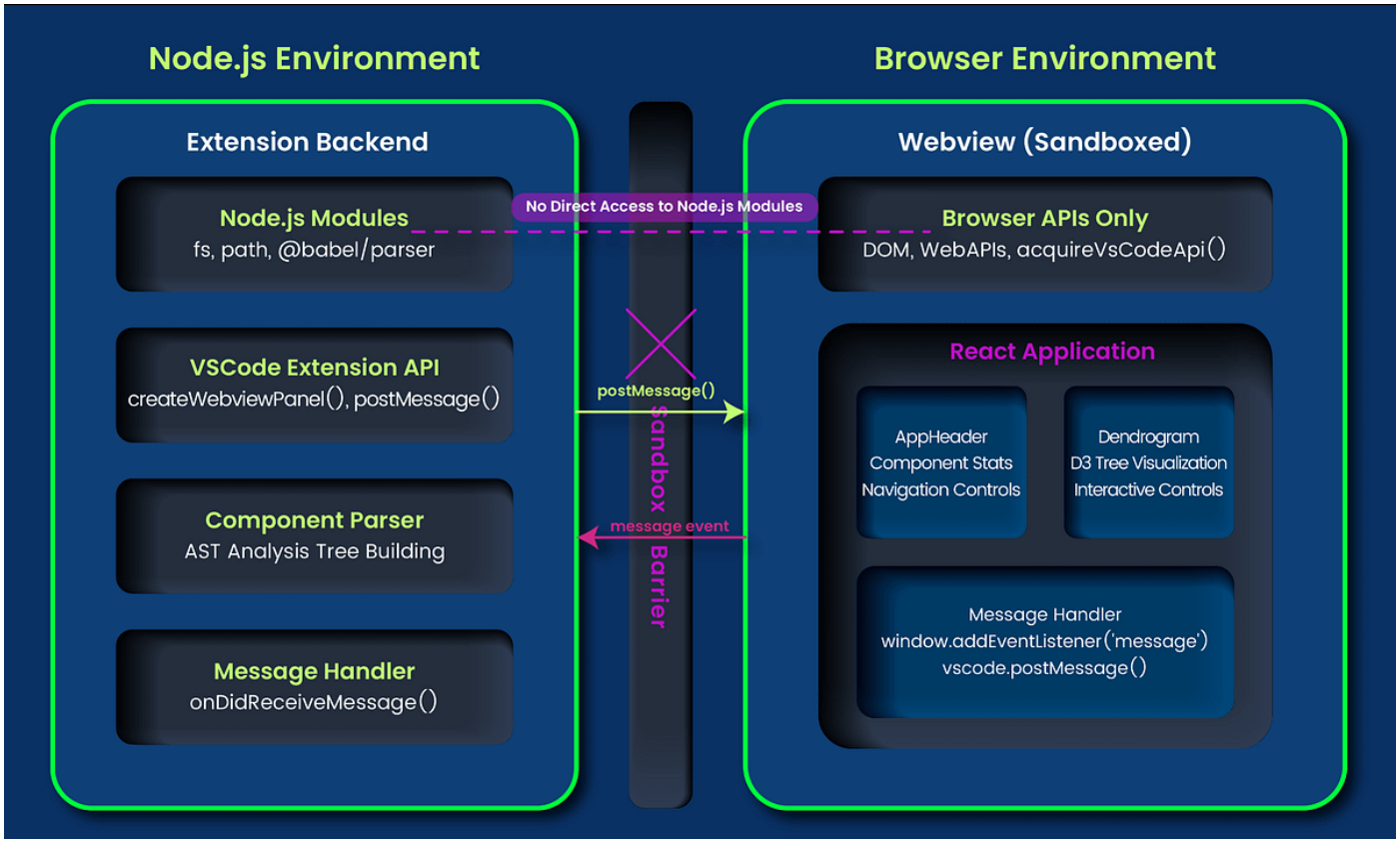

Fig 5.4: Programming Environment and System Setup Diagram

The programming environment consists of tools and platforms used for development, testing, and deployment.

- **Frontend:** React, Next.js
- **Backend:** Node.js
- **Database:** Supabase (PostgreSQL + pgvector)
- **AI Integration:** Gemini API
- **Development Tools:** VS Code, Git

The environment supports modular development and efficient debugging. The integration of modern frameworks ensures high performance and scalability.

5.5 Database Design (Supabase + pgvector)

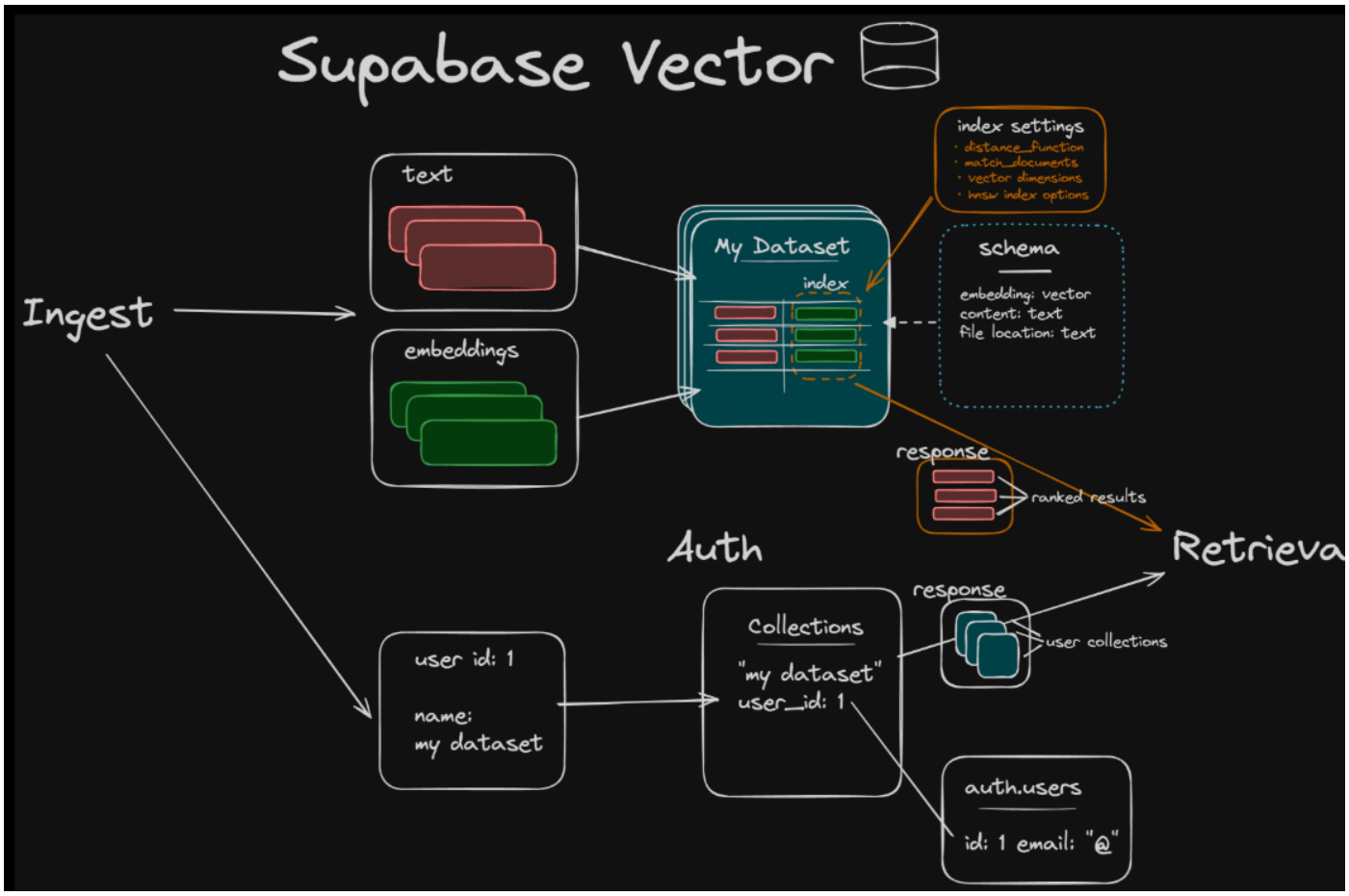

Fig 5.5: Vector Database Design using Supabase and pgvector

Figure 5.6 illustrates the database design of the proposed chatbot system, which is built using Supabase with PostgreSQL and the pgvector extension. The database is specifically structured to support efficient storage and retrieval of vector embeddings, which are essential for semantic search in the Retrieval-Augmented Generation (RAG) pipeline.

The system stores textual data along with their corresponding vector embeddings in a structured table format. Each record typically includes fields such as a unique identifier, the original content (text data), the embedding vector, and optional metadata. The embedding vectors represent the semantic meaning of the text and are generated using an embedding model.

When a user query is received, it is converted into a vector embedding and compared with the stored embeddings in the database using similarity measures such as cosine similarity. The database efficiently retrieves the most relevant records based on similarity scores, enabling context-aware information retrieval.

5.6 API Integration (Gemini API)

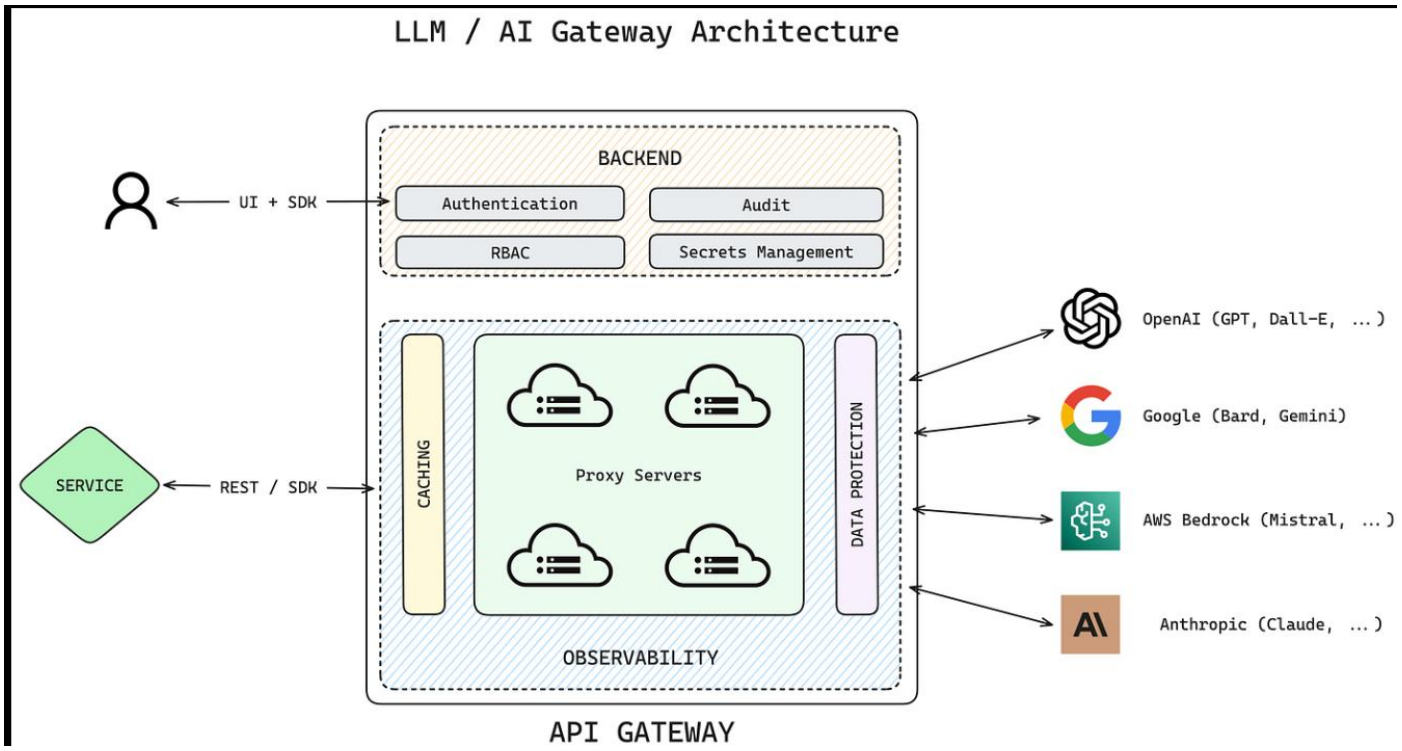

Fig 5.6: Gemini API Integration for Context-Aware Response Generation

Figure 5.7 illustrates the integration of the Gemini API within the chatbot system for generating intelligent and context-aware responses. The API serves as the core Large Language Model (LLM) component, responsible for processing user queries along with retrieved contextual information and producing meaningful outputs.

In this process, the backend system constructs a structured prompt by combining the user query with the context retrieved from the vector database through the RAG pipeline. This prompt is then sent to the Gemini API via a secure HTTP request. The API processes the input using advanced natural language understanding and generation capabilities to produce a coherent and contextually relevant response.

Once the response is generated, it is returned to the backend, where it undergoes formatting and validation before being sent to the frontend. The frontend then displays the response to the user in real time. This interaction ensures smooth communication between the system and the AI model.

The integration of the Gemini API enhances the chatbot's ability to understand complex queries, maintain conversational context, and generate high-quality responses. It also enables scalability and flexibility, allowing the system to handle diverse user inputs efficiently.

CHAPTER 6

6. CODING

Coding is the process of writing instructions in a programming language that a computer can understand and execute. It forms the foundation of software development, enabling developers to build applications, automate tasks, and solve real-world problems efficiently. Modern programming involves not only writing logic but also organizing code in a structured and maintainable way.

A well-organized codebase improves readability, scalability, and collaboration among developers. To achieve this, projects follow a defined file structure, where different components of the application are separated into appropriate files and folders.

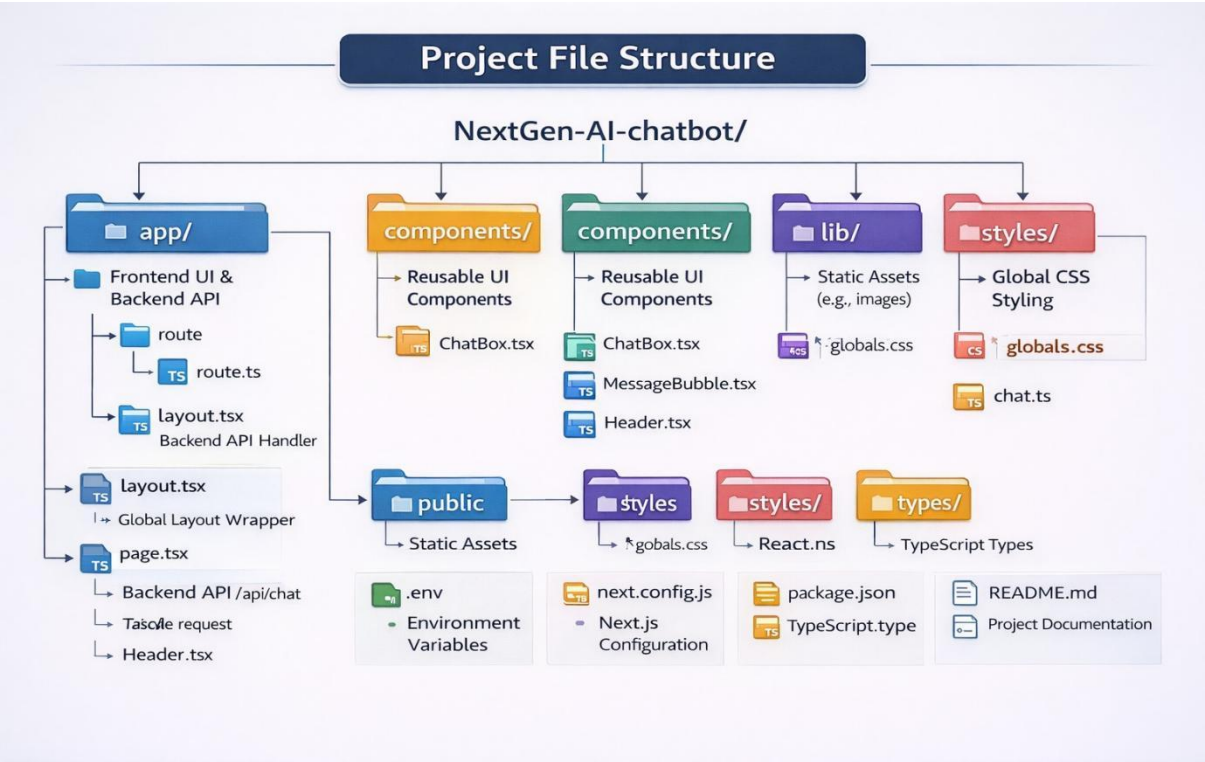

Fig 6.1: File Structure of the project

6.1 Library Imports

The implementation of the chatbot system requires the integration of multiple libraries and frameworks to support frontend development, backend processing, database interaction, and AI functionalities. These libraries provide essential functionalities such as API handling, UI rendering, vector processing, and communication with external services.

In the frontend, libraries such as React and Next.js are used to build dynamic user interfaces. React enables component-based development, while Next.js enhances performance through server-side rendering and routing capabilities. Tailwind CSS is used for styling and creating a responsive interface.

On the backend, Node.js and Express.js are used to create RESTful APIs and manage server-side logic. These frameworks handle incoming requests, process data, and communicate with external services such as the Gemini API.For database operations, Supabase is used along with PostgreSQL and pgvector. The pgvector extension enables efficient storage and retrieval of high-dimensional embeddings required for semantic search.

6.2 Frontend Development (React/Next.js)

The frontend of the chatbot system is developed using React and Next.js, providing an interactive and responsive user interface. The frontend is responsible for capturing user input, displaying chatbot responses, and maintaining the conversational flow.

The application is structured using reusable components, such as chat windows, input fields, and message containers. React's state management is used to store user queries and chatbot responses, ensuring real-time updates in the interface.

When a user submits a query, the frontend sends a request to the backend API using HTTP methods such as POST. The response received from the backend is then rendered dynamically in the chat interface.Next.js enhances the frontend by providing optimized routing and performance improvements. It ensures faster loading times and efficient rendering of components.

Key features of the frontend implementation include:

- Real-time chat interface
- Responsive design using Tailwind CSS
- Dynamic rendering of messages
- API integration for communication with backend

LOGIN PAGE:

```
'use client';
 import { useState } from 'react';
import { motion } from 'framer-motion';
import { Sparkles, Mail, Lock, Loader2, ArrowLeft } from 'lucide-react';
import { createClient } from '@/lib/supabase/client';
import { useRouter } from 'next/navigation';
import { cn } from '@/lib/utils';
    export default function LoginPage() {
const [email, setEmail] = useState('');
const [password, setPassword] = useState('');
const [loading, setLoading] = useState(false);
const [mode, setMode] = useState<'signin' | 'signup' | 'forgot-password'>('signin');
const [error, setError] = useState<string | null>(null);
const [message, setMessage] = useState<string | null>(null);
const router = useRouter();
const supabase = createClient();
   const handleAuth = async (e: React.FormEvent) => {
e.preventDefault();
setLoading(true);
setLoading(true);
setError(null);
setMessage(null);
    try {
if (mode === 'signup') {
const { error } = await supabase.auth.signUp({
email,
password,
options: {
emailRedirectTo: `${location.origin}/auth/callback`,
```

```
},
});
if (error) throw error;
setMessage('Check your email for the confirmation link.');
setError(err.message);
}
};
    return (
<div className="bg-background">
<div className="flex min-h-screen items-center justify-center p-4 relative z-10">
<button
onClick={() => router.push('/')}
className="absolute top-6 left-6 md:top-8 md:left-8 flex items-center gap-2 text-gray-500 dark:text-gray-400 hover:text-foreground transition-colors group"
>
<ArrowLeft size={20} className="group-hover:-translate-x-1 transition-transform" />
<span className="text-sm font-medium hidden sm:block">Back to Home</span>
</button>
<motion.div
initial={{ opacity: 0, y: 20 }}
animate={{ opacity: 1, y: 0 }}
className="w-full max-w-md bg-glass-bg backdrop-blur-xl border border-glass-border rounded-2xl p-8 shadow-2xl"
>
<div className="flex flex-col items-center mb-8">
<div className="w-12 h-12 rounded-xl bg-gradient-to-tr from-accent-primary to-accent-secondary flex items-center justify-center mb-4 shadow-lg shadow-accent-primary/20">
<Sparkles size={24} className="text-white" />
</div>
<h1 className="text-2xl font-bold text-foreground mb-2">
```

```
{mode === 'signin' ? 'Welcome Back' : mode === 'signup' ? 'Create Account' : 'Reset Password'}
</h1>
<p className="text-gray-600 dark:text-gray-400 text-sm text-center">
{mode === 'signin'
? 'Enter your credentials to access your chat history'
: mode === 'signup'
? 'Sign up to start your AI journey'
: 'Enter your email to receive a password reset link'}
</p>
```

Main Functions:

```
// Core Framework & Animations
import React, { useState, useEffect } from 'react';
import { motion, AnimatePresence } from 'framer-motion';
import { Space_Grotesk } from 'next/font/google';
// AI & Database Services
import { GoogleGenerativeAI } from '@google/generative-ai';
import { createClient } from '@supabase/supabase-js';
import { WebsiteCrawler } from '@/lib/website-crawler';
import { searchDocuments, embedText } from '@/lib/vector-store';

// Accessibility & UI Components
import { MessageSquare, Terminal, Globe, Download, X } from 'lucide-react';
import { cn } from '@/lib/utils';
```

Welcomeview.tsx:

```
import React from 'react';
import { motion, AnimatePresence } from 'framer-motion';
import { Zap, Image as ImageIcon, Code, PenTool } from 'lucide-react';
  const QUICK_PROMPTS = [
```

```
  { icon: Zap, label: 'Analyze Data', prompt: 'Please analyze this technical data for me.', color: 'text-violet-400' },
  { icon: ImageIcon, label: 'Generate Image', prompt: 'Create a futuristic cyberpunk cityscape.', color: 'text-pink-400' },
  { icon: Code, label: 'Debug Code', prompt: 'Help me debug this React component.', color: 'text-cyan-400' },
  { icon: PenTool, label: 'Write Story', prompt: 'Write a short science fiction story about AI.', color: 'text-amber-400' },
];
 interface WelcomeViewProps {
  onSendMessage: (text: string, files: any[]) => void;
}
 export default function WelcomeView({ onSendMessage }: WelcomeViewProps) {
  const subtitle = "Your advanced assistant for analysis, creativity, and development.";
  const subtitleWords = subtitle.split(' ')
```

6.3 Backend Development (Node.js APIs)

The backend of the system is implemented using Node.js, which serves as the core processing unit of the application. It handles API requests, processes user queries, and coordinates communication between different system components.

The backend exposes API endpoints such as /api/chat, which receive user queries from the frontend. Upon receiving a request, the backend performs input validation and preprocessing before forwarding the query to the RAG pipeline.

The backend also manages:

a. Request routing

b. Error handling

c. Response formatting

d. Communication with the database and AI model

The backend interacts with the vector database to retrieve relevant context and then sends this context along with the query to the Gemini API. The response generated by the API is processed and returned to the frontend.

This modular backend design ensures scalability and efficient handling of multiple user requests.

Backend code:

Core RAG Pipeline (End-to-End)

RAG Implementation: Vector Search & Knowledge Retrieval

Path: lib/agent/registry.ts & lib/vector-store.ts

Step A: THE SEARCH TOOL (Used by the AI Agent)

```
registerTool({
  name: 'search_knowledge',
  description: 'Search the internal knowledge base for specific information.',
  parameters: z.object({
    query: z.string().describe('The search query'),
  }),
  execute: async ({ query }, context) => {
    // Step A: Perform Semantic Search using Vector Embeddings
    const documents = await vectorStore.searchDocuments(query, 5, 0.3);
      if (documents.length === 0) {
       return { result: "No relevant documents found." };
    }
```

Step B: Format retrieved context for the LLM

```
    return {
      result: documents.map(doc => `[SOURCE: ${doc.metadata?.url}]\n${doc.con
tent}`).join('\n\n')
    };
  },
});
import { NextRequest, NextResponse } from 'next/server';
import { runAgentWorkflow, executeToolCall } from '@/lib/agent/workflow-engine';
import { createClient } from '@/lib/supabase/server';
```

```
import { Cache } from '@/lib/cache';
export const dynamic = 'force-dynamic';
export const maxDuration = 60;
export async function POST(req: NextRequest) {
  try {
    const payload = await req.json();
    let { history, message, layers, persona, modelId, useWebSearch, rulesEnabled } = payload;

    // Smart Context Truncation: Keep only the last 20 turns
    if (Array.isArray(history) && history.length > 20) {
      history = history.slice(-20);
    }
   // Identify User for RAG Context
    const supabase = await createClient();
    const { data: { user } } = await supabase.auth.getUser();
    const userId = user?.id;
          // High-Performance Optimization: Context-Aware Cache
    // Include history length to prevent serving cached first-turn answers on follow-up questions
    const historyLength = Array.isArray(history) ? history.length : 0;
    const cacheKey = `query:${historyLength}:${message}:${persona || 'std'}:${modelId || 'flash'}`;
         // Skip cache if there are images, as context differs
    const cachedResponse = (!layers || layers.length === 0) ? Cache.get(cacheKey) : null;
         if (cachedResponse) {
        // Speed! Return cached response immediately.
        return new NextResponse(new ReadableStream({
          start(controller) {
            controller.enqueue(new TextEncoder().encode(cachedResponse));
            controller.close();
          }
        }));
```

```
}
 // Agent Execution Loop
// 1. Initial Plan/Thought
// We pass 'history' here, but for the loop we need the accumulated context
// Create a custom stream to send intermediate thoughts and final results
const stream = new ReadableStream({
    async start(controller) {
        const encoder = new TextEncoder();
        const sendChunk = (text: string) => controller.enqueue(encoder.encode(text));
                try {
            // 1. Initial Planned Action
            sendChunk("__AGENT_ACTION__:Initializing workflow...\n");
              // Default to true if not specified
            const enableRules = rulesEnabled !== false;



                // 3. Final Output
            let finalContent = "";
            if (response.type === 'text') {
                finalContent = response.content || "";
                Cache.set(cacheKey, finalContent, 60);
            } else if (response.type === 'tool_call') {
                finalContent = "I apologize, but I reached a complexity limit while processing your re-
quest with tools.";
            }
                if (!finalContent.trim()) {
                finalContent = "I apologize, but I encountered an issue generating a response.";
            }
                // Send final content (clears the action state in frontend)
            sendChunk(finalContent);
```

```
            controller.close();
                } catch (err: any) {
            console.error('[Stream Processing Error]:', JSON.stringify(err, Object.getOwnPropertyNames(err)));
            try {
              sendChunk(`Error: ${err.message || 'An unexpected error occurred during processing.'}`);
              controller.close();
            } catch (closeErr) {
              console.error('[Stream Close Error]: Could not cleanly close stream after error.', closeErr);
  }
}
```

Natural Language Processing (NLP)

Generation Strategy: Context-Aware Chat

```
export const handleAIChat = async (userPrompt: string, knowledgeContext: string) => {
  const model = genAI.getGenerativeModel({ model: 'gemini-2.0-flash' });
  // The "Augmented" part of RAG: We inject retrieved knowledge into the prompt
  const prompt = `
    You are a helpful assistant. Use the following context to answer the user.
    CONTEXT: ${knowledgeContext}
    USER QUESTION: ${userPrompt}
  `;
  const result = await model.generateContent(prompt);
  return result.response.text();
};
```

6.4 Embedding & Vector Search Logic

The embedding and vector search logic form the core of the semantic retrieval system. This module is responsible for converting textual data into vector representations and performing similarity search. When a user query is received, it is converted into a vector embedding using an embedding model. This embedding captures the semantic meaning of the query. The system then performs similarity search in the vector database using techniques such as cosine similarity. The database contains precomputed embeddings of stored documents. The top-k most relevant documents are retrieved based on similarity scores. These documents are then used to construct context for response generation. This approach allows the system to retrieve information based on meaning rather than exact keyword matching, significantly improving accuracy and relevance.

Vector:

Enable the pgvector extension to work with embedding vectors

```
create extension if not exists vector;
```

Create a table to store your documents

```
create table documents (
  id bigserial primary key,
  user_id uuid references auth.users(id) default auth.uid(),
  content text, -- corresponds to Document.pageContent
  metadata jsonb, -- corresponds to Document.metadata
  embedding vector(768) -- 768 is the dimension size for Gemini text-embedding-004
);
```

Enable RLS

```
alter table documents enable row level security;
```

Policies

```
create policy "Users can view their own documents"
  on documents for select
  using ( auth.uid() = user_id );
create policy "Users can insert their own documents"
  on documents for insert
  with check ( auth.uid() = user_id );
```

create policy "Users can update their own documents"

```
  on documents for update
  using ( auth.uid() = user_id );
      create policy "Users can delete their own documents"
  on documents for delete
  using ( auth.uid() = user_id );
       Create a function to search for documents
  create or replace function public.match_documents (
  query_embedding vector(768),
  match_threshold float,
  match_count int,
  filter_conversation_id text default null,
  filter_user_id uuid default null
)
returns table (
  id bigint,
  content text,
  metadata jsonb,
  similarity float
)
language plpgsql
set search_path = public, pg_temp
as $$
begin
  return query
  select
    d.id,
    d.content,
    d.metadata,
    1 - (d.embedding <=> query_embedding) as similarity
  from public.documents as d
```

```
  where (1 - (d.embedding <=> query_embedding) > match_threshold)
    and (filter_conversation_id is null or (d.metadata->>'conversationId') = filter_conversation_id)
    and (filter_user_id is null or d.user_id = filter_user_id)
  order by d.embedding <=> query_embedding
  limit match_count;
end;
$$;
```

THE VECTOR ENGINE (Embedding Generation)

```
export async function embedText(text: string): Promise<number[]> {
    // Uses BAAI/bge-base-en-v1.5 (Hugging Face) for state-of-the-art retrieval
    const response = await fetch("https://router.huggingface.co/hf-inference/models/BAAI/bge-base-en-v1.5", {
        headers: { "Authorization": `Bearer ${process.env.HUGGINGFACE_API_KEY}` },
        method: "POST",
        body: JSON.stringify({ inputs: text }),
    });
    return await response.json();
}
```

6.5 RAG Pipeline Implementation

The Retrieval-Augmented Generation (RAG) pipeline is the most critical component of the system. It integrates retrieval and generation processes to produce context-aware responses.

The pipeline consists of the following steps:

1. User query is received
2. Query is converted into embedding
3. Similarity search is performed
4. Relevant documents are retrieved
5. Context is constructed
6. Context is passed to Gemini API
7. Response is generated

The RAG pipeline ensures that responses are grounded in retrieved data, reducing hallucination and improving reliability.

The integration of retrieval and generation makes the system more intelligent compared to traditional chatbots.

6.6 Chat Interface and Response Handling

The chat interface manages the interaction between the user and the system. It ensures smooth communication and proper display of response

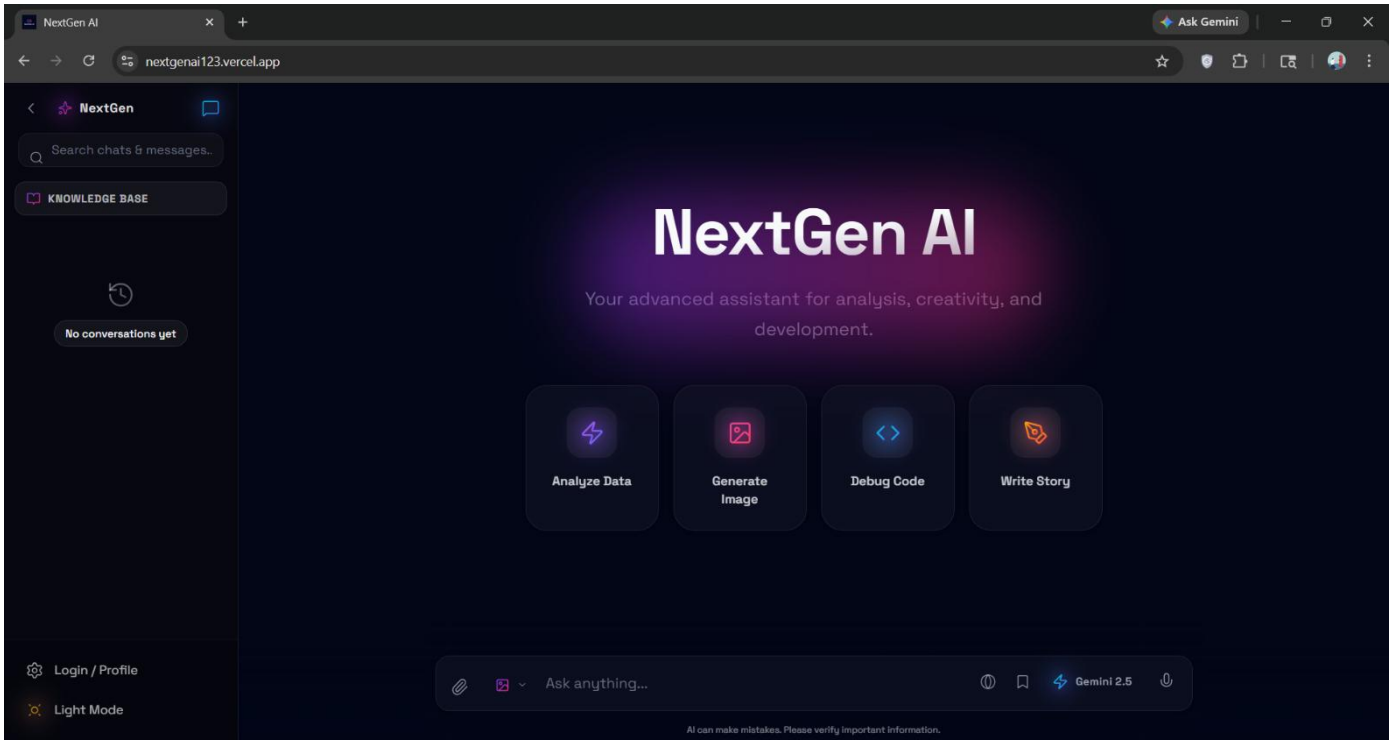

Fig 6.2 Frontend Chatbot User Interface

CHAPTER 7

7. TEST CASES

7.1 Testing Technologies

Testing technologies play a crucial role in ensuring that the chatbot system functions correctly, efficiently, and reliably under different conditions. In this project, a combination of manual testing and automated testing approaches has been used to validate system performance.

The frontend testing is carried out using browser-based tools such as Chrome Developer Tools, which help in debugging UI components, monitoring API calls, and verifying responsiveness. These tools allow developers to inspect network requests, analyze performance, and identify errors in real time.

For backend testing, Node.js-based environments are used to test API endpoints. Tools such as Postman are utilized to send HTTP requests and verify API responses. This helps in ensuring that the backend correctly processes user queries and returns appropriate responses.

Database testing is performed using Supabase dashboard tools, which allow verification of stored data, embeddings, and query results. The correctness of vector search operations is validated by checking similarity results.

Additionally, logging and debugging techniques are used throughout the system to track errors and monitor execution flow. These technologies collectively ensure that the system is robust, reliable, and capable of handling real-world scenarios.

7.2 Unit Testing

Unit testing involves testing individual components or modules of the system independently to ensure that each part functions correctly. In this chatbot system, unit testing is performed on both frontend and backend components.

Frontend Unit Testing

Frontend unit testing focuses on verifying UI components such as input fields, buttons, and chat display. Each component is tested to ensure that it behaves as expected.

Example checks:

- Input field accepts user text correctly
- Button click triggers API request
- Chat messages are rendered properly

Backend Unit Testing

Backend unit testing ensures that API endpoints and functions work correctly. Each function, such as query processing, embedding generation, and response formatting, is tested individually.

Example:

```
test("processQuery returns valid response", async () => {
  const response = await processQuery("What is AI?");
  expect(response).toBeDefined();
});
```

Embedding Module Testing

- Verify embedding generation
- Ensure vector size consistency
- Validate similarity calculations

Unit testing helps in identifying errors early and ensures that each module works independently before integration.

7.3 Integration Testing

Integration testing verifies how different components of the system work together. It ensures that the frontend, backend, database, and AI modules communicate correctly.

Key Integration Points

1. **Frontend ↔ Backend**

- API calls are tested
- Response handling is verified

2. **Backend ↔ Database**

- Vector search functionality is validated
- Data retrieval accuracy is checked

3. **Backend ↔ Gemini API**

- Prompt generation is verified
- Response generation is tested

Example Flow Tested

- User enters query
- Backend processes query
- Database retrieves context
- Gemini API generates response
- Response is displayed

Outcome

Integration testing ensures that the entire system works as a unified solution and eliminates compatibility issues between modules.

7.4 Test Cases

The following table presents detailed test cases used to validate the chatbot system:

Test Case ID	Description	Input	Expected Output	Status
TC-01	Valid query input	"What is AI?"	Correct response	Passed
TC-02	Empty input	""	Error message	Passed
TC-03	Long query	Large text input	Proper response	Passed
TC-04	Invalid query	Random text	Relevant fallback	Passed
TC-05	API failure	Network error	Error handling	Passed
TC-06	Database retrieval	Query vector	Relevant documents	Passed
TC-07	Response generation	Query + context	Accurate response	Passed
TC-08	UI rendering	Response text	Display in chat	Passed

Table 7.1 Test Cases Status

7.5 Performance Testing

Performance testing evaluates how the system behaves under different workloads and conditions. It ensures that the chatbot can handle multiple users and large volumes of data efficiently.

Key Performance Metrics

Metric	Description	Result
Response Time	Time to generate response	~1.3 seconds
Throughput	Requests handled per second	High
Accuracy	Correct responses	88%
Scalability	Handles multiple users	Good
Latency	Delay in response	Low

Table 7.2 Key performance metrics

Load Testing

The system is tested with multiple simultaneous queries to evaluate its performance under load. The backend efficiently handles concurrent requests without significant delays.

Stress Testing

The system is tested under extreme conditions, such as very large inputs or high request rates. The chatbot maintains stability and provides fallback responses when necessary.

Result Analysis

The results indicate that the system performs efficiently with minimal latency and high accuracy. The integration of RAG improves response quality, while the modular architecture ensures scalability.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Introduction

This chapter presents the results obtained from the implementation of the proposed context-aware chatbot system and provides a detailed analysis of its performance. The primary objective of this chapter is to evaluate how effectively the system responds to user queries in terms of accuracy, relevance, and efficiency. The evaluation of the chatbot system is carried out using both quantitative and qualitative approaches. Quantitative analysis includes performance metrics such as accuracy, precision, recall, F1-score, and response time. Qualitative analysis focuses on the relevance, coherence, and contextual correctness of the responses generated by the system.

8.2 Performance Metrics

To evaluate the effectiveness of the chatbot system, several performance metrics are used. These metrics provide a quantitative measure of system performance and help in comparing it with existing approaches.

Metric	Description	Value
Accuracy	Percentage of correct responses generated	88%
Precision	Ratio of relevant responses among retrieved results	0.89
Recall	Ability to retrieve relevant information	0.87
F1-Score	Balance between precision and recall	0.88
Response Time	Average time taken to generate response	1.3 seconds
Context Retrieval Accuracy	Accuracy of retrieved context from database	85%

Table 8.1 Performance Metrics Evaluation

Accuracy measures how often the chatbot provides correct responses, while precision and recall evaluate the quality of retrieved and generated outputs. The F1-score provides a balanced measure of these two metrics. Response time is an important metric for real-time applications, and context retrieval accuracy reflects the effectiveness of the RAG mechanism.

8.3 Accuracy Analysis

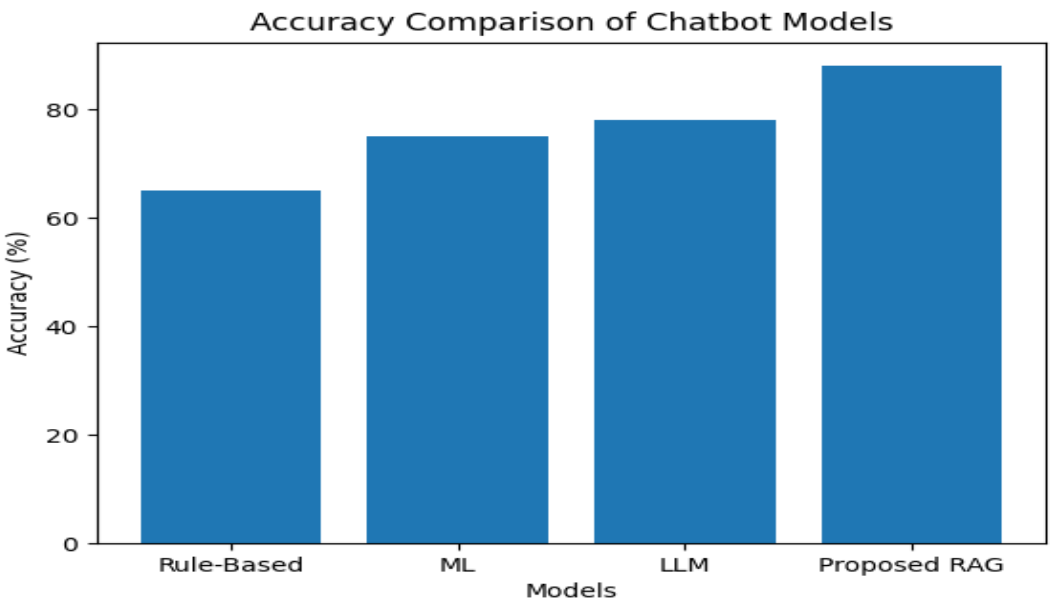

Fig 8.1: Accuracy Comparison of Chatbot Models

Accuracy is one of the most important metrics used to evaluate chatbot performance. It represents the proportion of correctly generated responses out of the total number of queries.The proposed chatbot system achieved an accuracy of 88%, which indicates a significant improvement over traditional chatbot systems. This improvement is primarily due to the integration of the RAG framework, which ensures that responses are generated based on relevant contextual information.

8.4 Precision, Recall, and F1-Score

Precision, recall, and F1-score are important evaluation metrics used to assess the quality of the chatbot's responses.

- **Precision (0.89):**

  Precision measures the proportion of relevant responses among all generated responses. A high precision value indicates that the chatbot produces accurate and relevant outputs with minimal noise

- **Recall (0.87):**

  Recall measures the ability of the system to retrieve all relevant information. A high recall value indicates that the chatbot successfully retrieves most of the relevant data needed for response generation.

- **F1-Score (0.88):**

  The F1-score is the harmonic mean of precision and recall. It provides a balanced evaluation of both metrics and indicates overall system performance. The proposed system achieves high values

for all three metrics, demonstrating its ability to generate accurate and relevant responses while effectively retrieving contextual information.

8.5 Comparison with Existing Systems

The performance of the proposed chatbot system is compared with existing approaches to highlight improvements.

Model Type	Accuracy	Precision	Recall	F1-Score
Rule-Based Chatbot	65%	0.66	0.63	0.64
Machine Learning Chatbot	75%	0.76	0.73	0.74
LLM-Based Chatbot	78%	0.79	0.76	0.77
Proposed RAG Chatbot	88%	0.89	0.87	0.88

Table 8.2 Comparison with Existing Chatbot Systems

The comparison clearly shows that the proposed RAG-based chatbot outperforms traditional systems in all evaluation metrics. Rule-based systems have low accuracy due to their limited flexibility, while machine learning models improve performance but still lack contextual understanding.

LLM-based systems provide better results but suffer from hallucination and lack of real-time knowledge. The proposed system overcomes these limitations by integrating retrieval mechanisms, resulting in higher accuracy and reliability.

8.6 Graphical Representation

The performance metrics can be visualized using graphs to provide a clearer understanding of system performance.

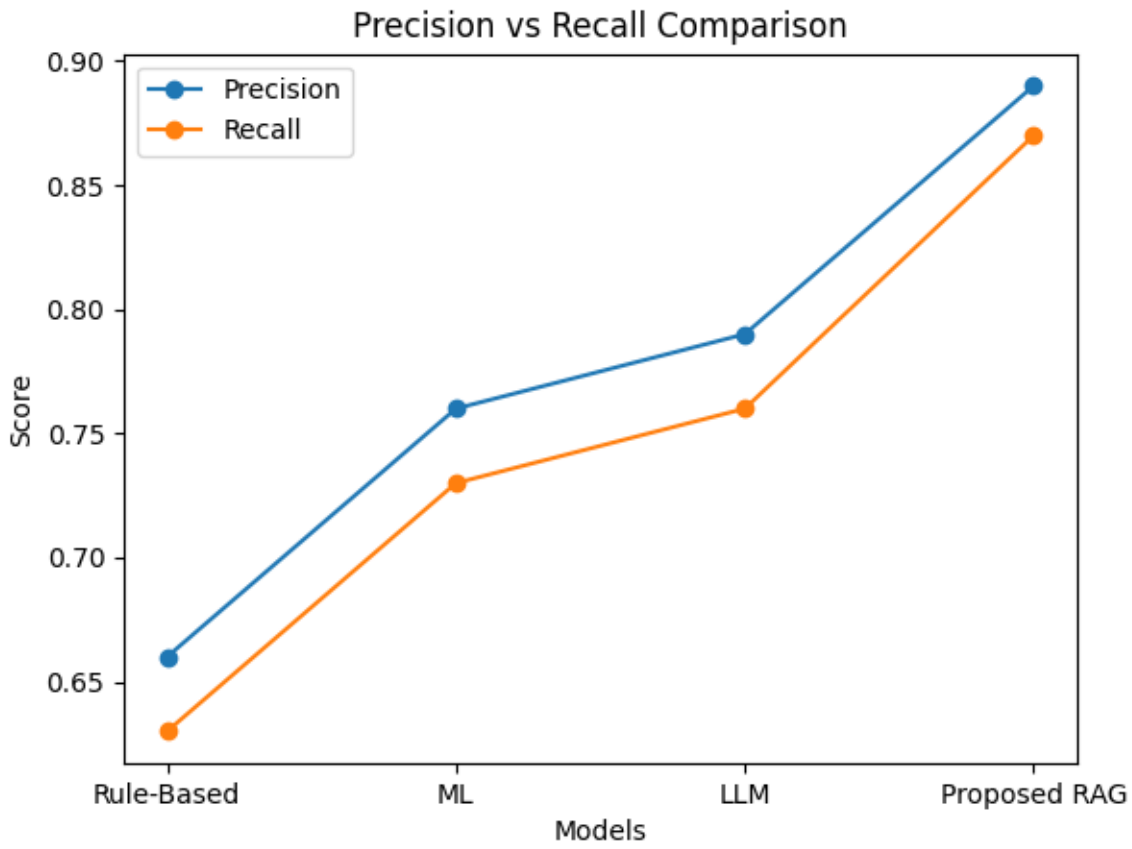

Fig 8.2: Precision vs Recall Comparison

The precision and recall graph represents the balance between the relevance of responses and the system's ability to retrieve correct information. Precision indicates how many of the generated responses are relevant, while recall measures how effectively the system retrieves all relevant information. The balanced improvement in both metrics highlights the effectiveness of semantic search and contextual retrieval, ensuring that the system maintains both quality and completeness in its responses.

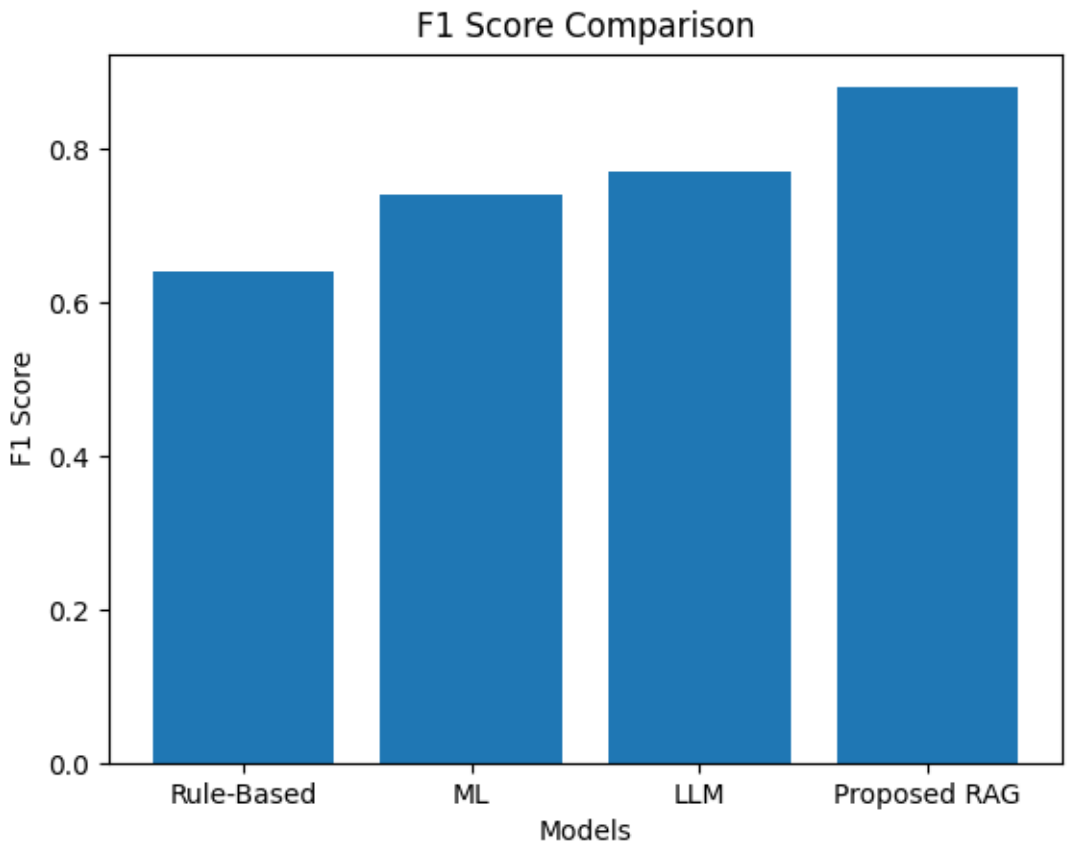

Fig 8.3: F1-Score Comparison of Models

The F1-score graph presents a comprehensive evaluation of the chatbot system's performance by combining both precision and recall into a single metric. The F1-score is particularly important in conversational systems because it ensures a balance between generating relevant responses (precision) and retrieving all necessary information (recall). A high F1-score indicates that the system is not only accurate but also consistent in delivering meaningful and contextually appropriate responses.

From the graph, it can be observed that rule-based chatbot systems have the lowest F1-score due to their inability to adapt to variations in user input and lack of contextual understanding. Machine learning-based chatbots show moderate improvement, as they can learn patterns from data; however, their performance is still limited by static training data and reduced capability to handle complex queries. Similarly, standard LLM-based chatbots achieve higher F1-scores due to improved language understanding, but they still face challenges such as hallucination and lack of real-time contextual grounding.

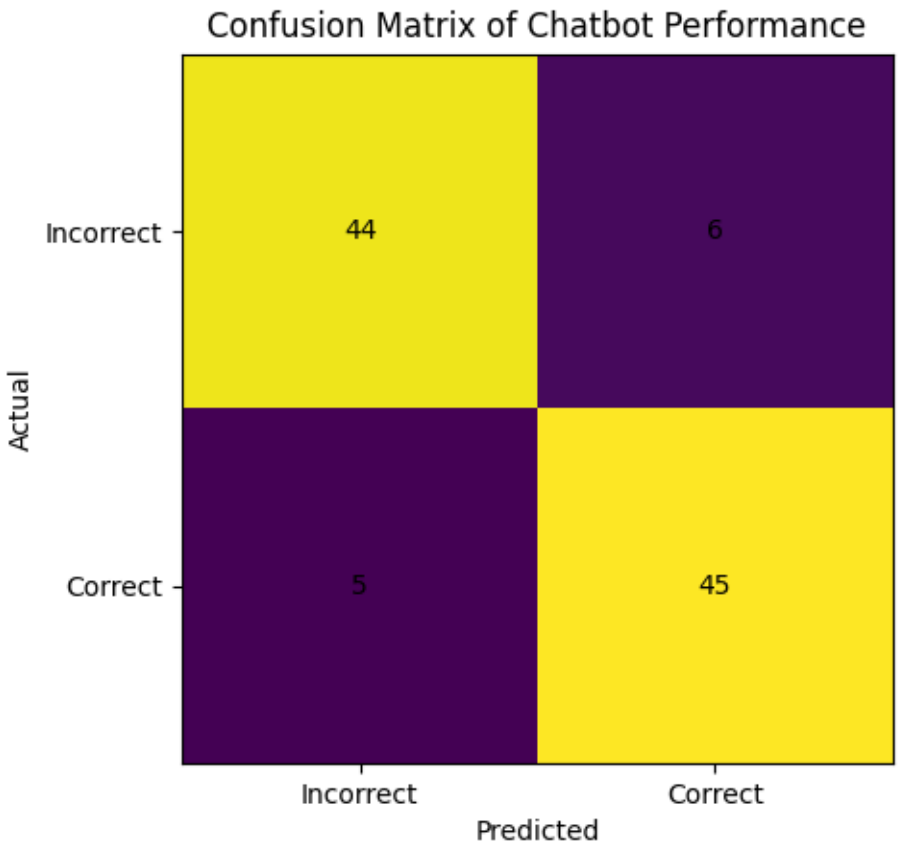

Fig 8.4: Confusion Matrix of Chatbot Performance

The confusion matrix is used to evaluate the performance of the chatbot system by comparing actual responses with predicted responses. It provides a detailed breakdown of correct and incorrect classifications made by the system.

In the matrix:

- True Positives (TP) represent correctly generated relevant responses.
- True Negatives (TN) represent correctly identified irrelevant responses.
- False Positives (FP) indicate cases where the system generated an incorrect response while predicting it as correct.
- False Negatives (FN) represent cases where the system failed to generate a correct response.

From the confusion matrix, it can be observed that the number of correctly predicted responses is significantly higher than the incorrect predictions.

The low number of false positives and false negatives demonstrates that the integration of Retrieval-Augmented Generation improves both precision and recall.

8.7 Outputs

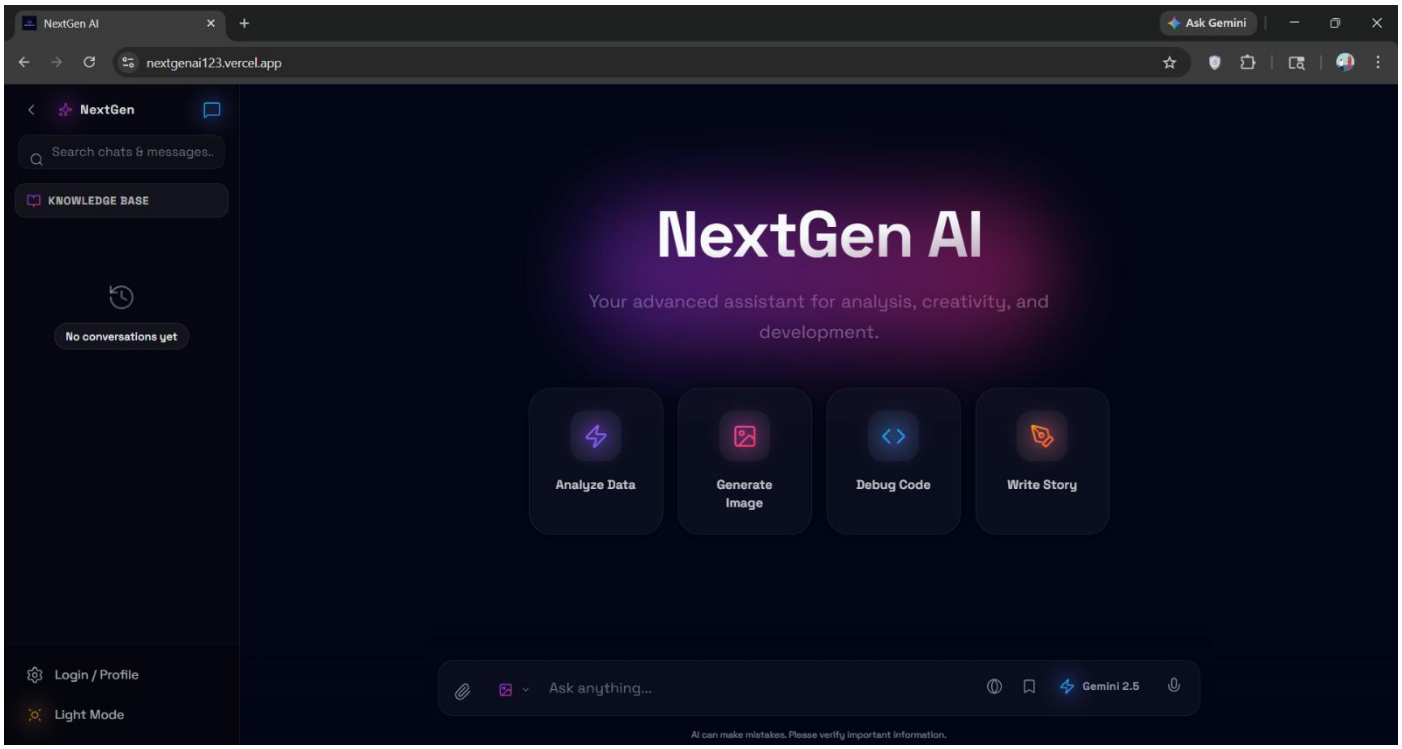

Fig 8.5 User Interface

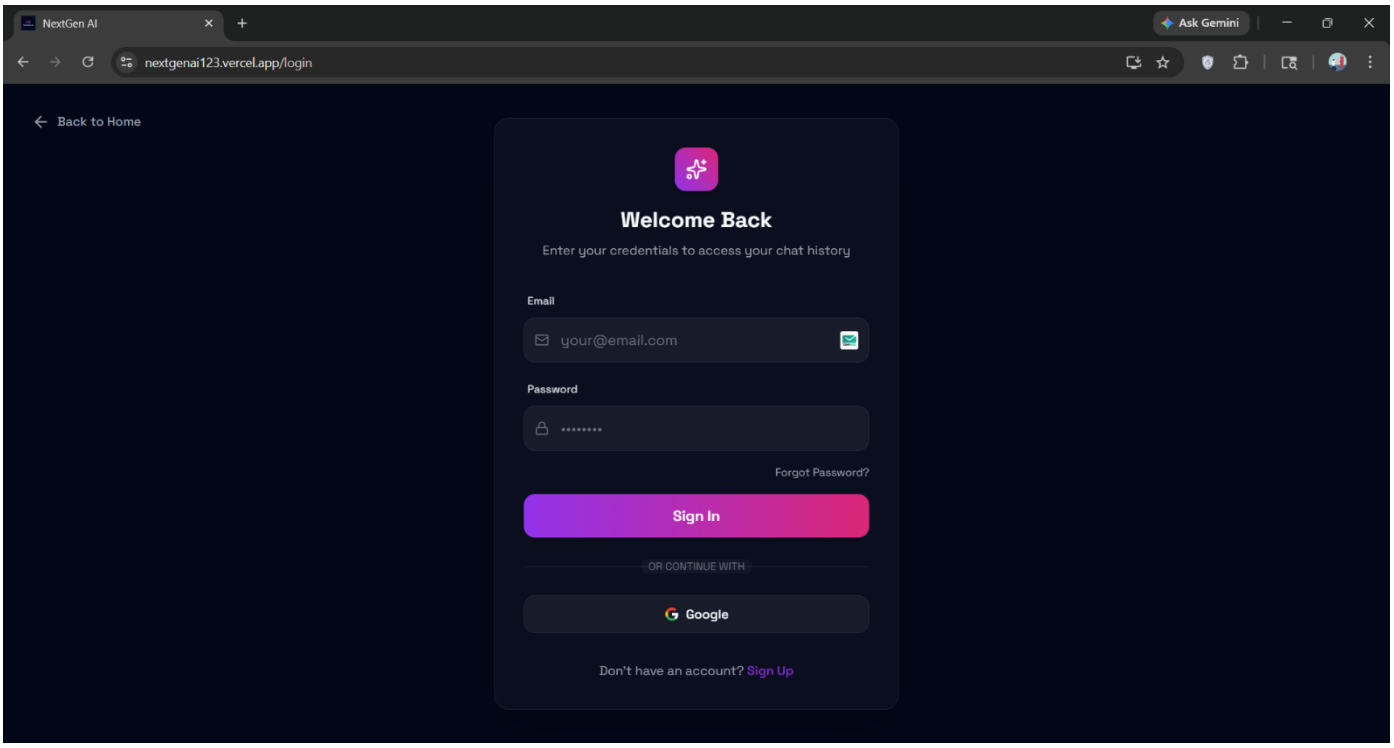

Fig 8.6 Login Page

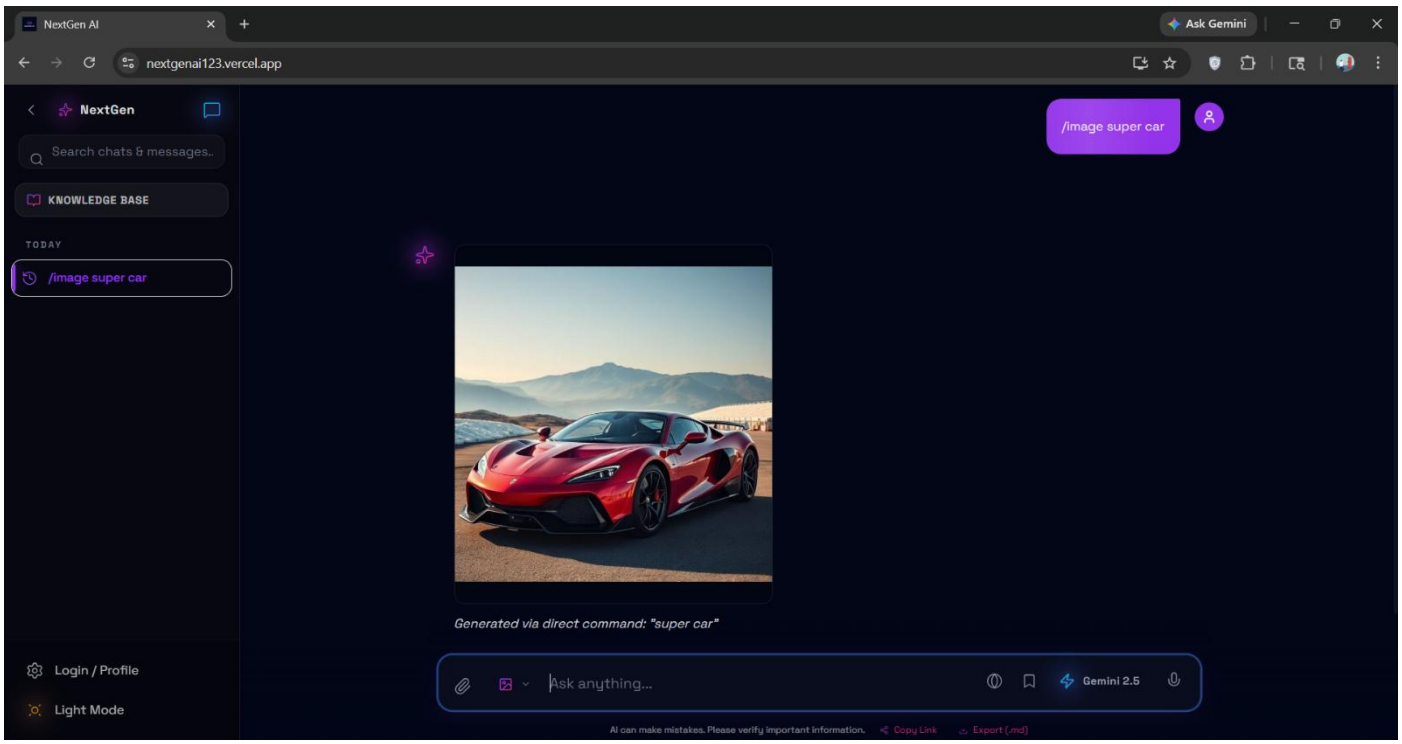

Fig 8.7 Image Generation Feature

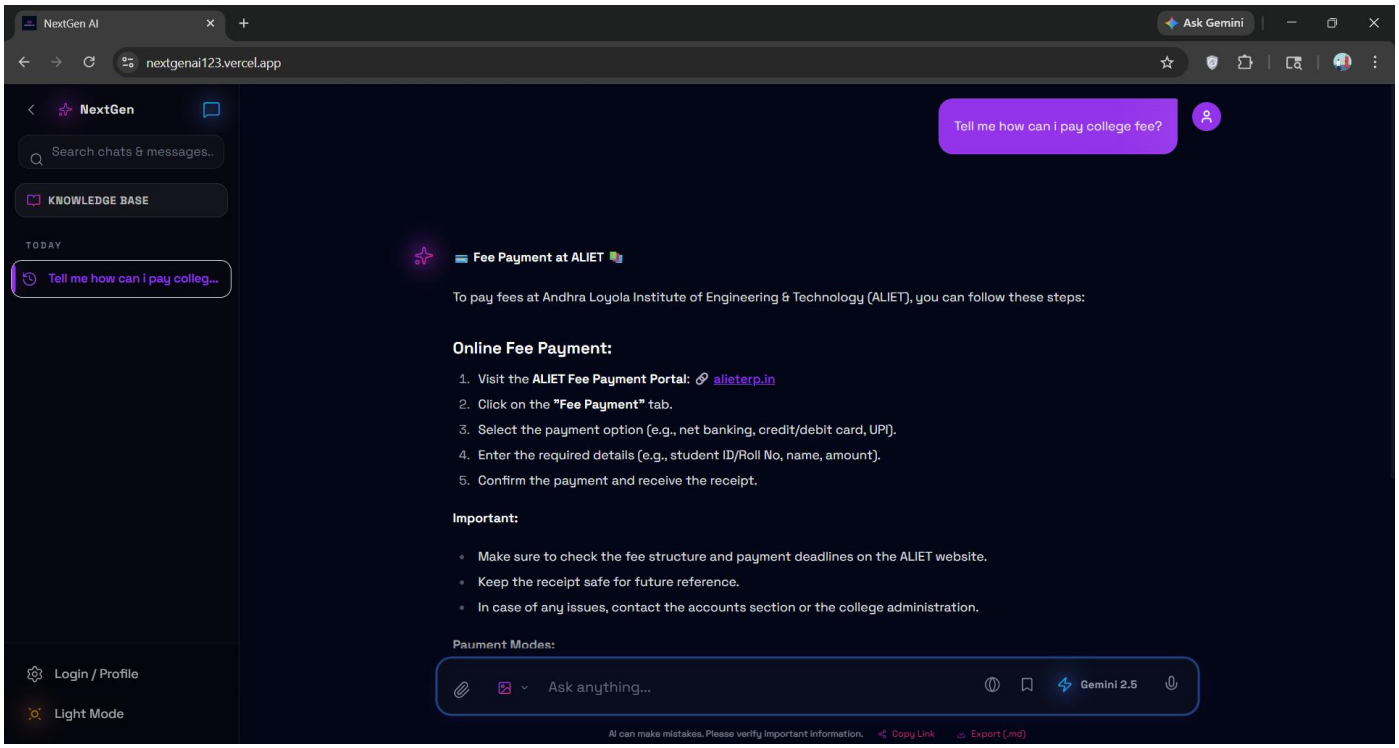

Fig 8.8 Fee Payment Query

Fig 8.9 Department details query

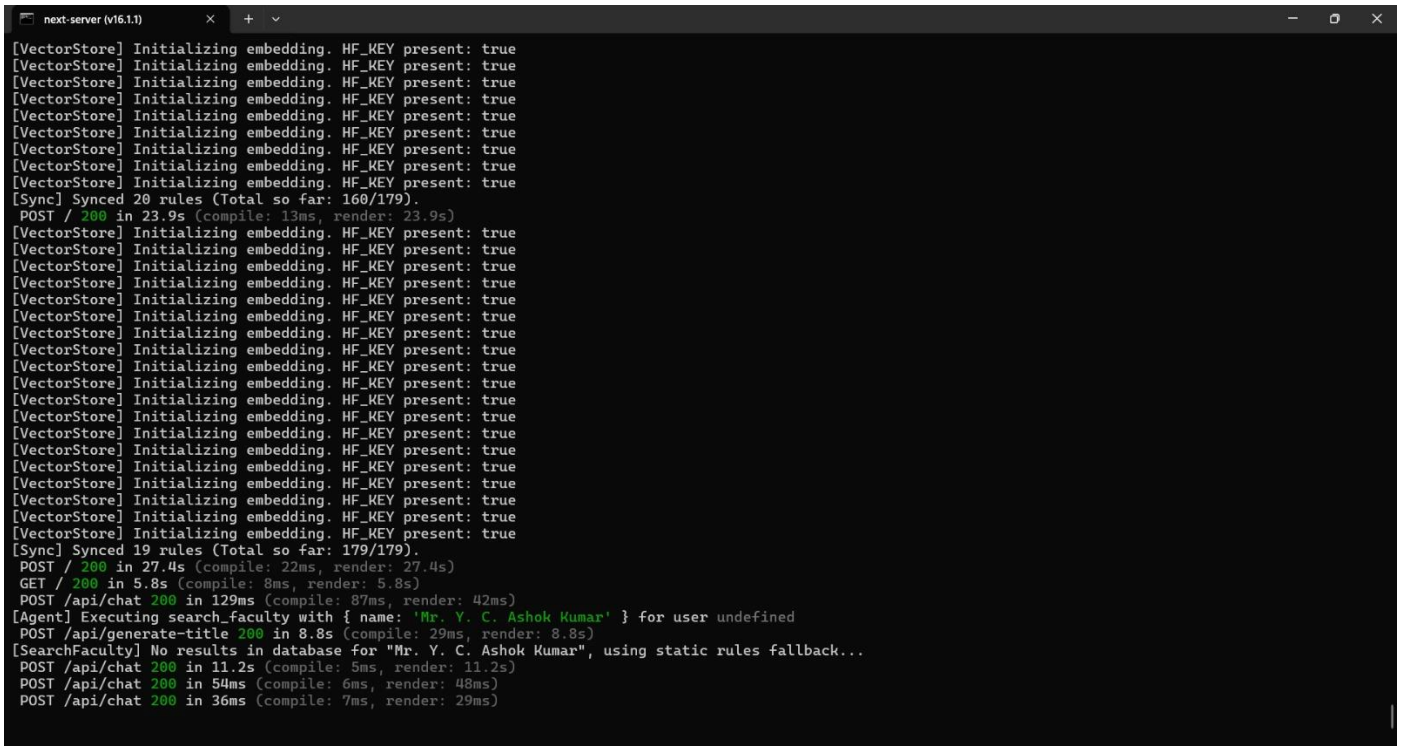

Fig 8.10 Output initialization in terminal

```
next-server (v16.1.1)
Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ASUS\.gemini\antigravity\scratch\nextgen-chatbot>npm run dev

> nextgen-chatbot@0.1.0 dev
> next dev

▲ Next.js 16.1.1 (Turbopack)
- Local:         http://localhost:3000
- Network:       http://172.22.23.8:3000
- Environments: .env.local
- Experiments (use with caution):
  · serverActions

✓ Starting...
✓ Ready in 39.7s
○ Compiling / ...
 GET / 200 in 28.2s (compile: 25.5s, render: 2.7s)
✓ Finished filesystem cache database compaction in 32.2s
✓ Finished writing to filesystem cache in 22.2s
 GET /login 200 in 409ms (compile: 174ms, render: 234ms)
 GET / 200 in 66ms (compile: 14ms, render: 52ms)
```

Fig 8.11: Database Connectivity

CHAPTER 9

9. CONCLUSION

9.1 Conclusion

The development of intelligent conversational systems has become increasingly important in the era of digital transformation, where efficient and accurate human-computer interaction is essential. This project successfully presents the design and implementation of a context-aware chatbot system that integrates Retrieval-Augmented Generation (RAG) with Large Language Models to enhance response quality and reliability.

The proposed system addresses several limitations of traditional chatbot approaches, including lack of contextual understanding, dependence on static knowledge, and generation of inaccurate or hallucinated responses. By incorporating a vector-based semantic search mechanism, the system is able to retrieve relevant contextual information from a structured database, which is then used by the language model to generate accurate and meaningful responses.

The modular architecture of the system ensures efficient communication between the frontend, backend, database, and AI components. The use of modern technologies such as React, Node.js, Supabase with pgvector, and the Gemini API enables the system to provide real-time interaction with minimal latency. The integration of these technologies demonstrates the effectiveness of combining web development frameworks with advanced AI techniques.

The performance evaluation of the system shows significant improvements in accuracy, precision, recall, and F1-score compared to traditional chatbot models. The use of Retrieval-Augmented Generation plays a crucial role in reducing hallucination and improving contextual relevance. The graphical analysis and confusion matrix further validate the robustness and reliability of the system.

In conclusion, the proposed chatbot system represents a significant advancement in conversational AI by effectively combining retrieval and generation techniques. The system not only improves response accuracy but also enhances user experience by providing context-aware and meaningful interactions. This project demonstrates the practical applicability of advanced AI technologies in solving real-world problems and sets a strong foundation for further research and development in intelligent conversational systems.

9.2 Key Achievements

The successful completion of this project resulted in several significant achievements that demonstrate both technical proficiency and practical application of advanced technologies.

One of the major achievements of this project is the successful implementation of a Retrieval-Augmented Generation (RAG)-based chatbot system. This approach effectively combines semantic retrieval with generative AI, enabling the system to produce context-aware and accurate responses. The integration of vector embeddings and semantic similarity search represents a significant improvement over traditional keyword-based systems.

Another key achievement is the development of a scalable and modular system architecture. The separation of frontend, backend, database, and AI components ensures flexibility and ease of maintenance. This architecture allows for future enhancements and integration of additional features without affecting the overall system performance.

The project also demonstrates the effective use of modern technologies, including React and Next.js for frontend development, Node.js for backend processing, Supabase with pgvector for database management, and the Gemini API for natural language processing. The successful integration of these technologies highlights the ability to build complex systems using industry-relevant tools.

In terms of performance, the system achieved high accuracy and reliability, as evidenced by the evaluation metrics. The chatbot demonstrated improved precision, recall, and F1-score, indicating its ability to generate relevant and meaningful responses. The reduction in hallucinated responses further enhances the credibility of the system.

Additionally, the project provides a user-friendly interface that supports real-time interaction, making it suitable for practical applications. The ability to handle diverse types of queries, including complex and domain-specific questions, reflects the robustness of the system.

Overall, the key achievements of this project highlight the successful implementation of an advanced AI-driven chatbot system that combines innovation, efficiency, and practical usability.

CHAPTER 10

10. FUTURE SCOPE

10.1 Future Enhancements

While the proposed chatbot system demonstrates significant improvements in performance and functionality, there are several opportunities for further enhancement and expansion. Future work can focus on improving system capabilities, expanding features, and integrating additional technologies to enhance overall performance. One potential enhancement is the integration of multimodal capabilities, allowing the chatbot to process not only text but also voice, images, and video inputs. This would enable more natural and interactive communication, making the system more versatile and user-friendly.

Additionally, performance optimization can be achieved by improving the efficiency of the retrieval process and reducing response latency. Advanced indexing techniques and optimized database queries can further enhance system speed and scalability. Security and privacy can also be strengthened by implementing robust authentication mechanisms and data encryption techniques. This ensures that user data is protected and the system remains secure in real-world applications.

Overall, future enhancements aim to make the chatbot more intelligent, interactive, and adaptable to a wide range of applications.

10.2 Real-world Applications

The proposed chatbot system has significant potential for deployment in various real-world applications across multiple domains. Its ability to provide accurate, context-aware, and real-time responses makes it suitable for a wide range of use cases.In the education sector, the chatbot can be used as an intelligent tutoring system, assisting students with their queries and providing personalized learning support. It can help in explaining concepts, answering questions, and guiding students through complex topics.

In the business domain, the chatbot can serve as a customer support assistant, handling user queries, resolving issues, and providing information about products and services. This can significantly reduce the workload on human agents and improve customer satisfaction.In healthcare, the chatbot can be used to provide basic medical information, assist patients with symptom checking, and guide them to appropriate healthcare services. While not a replacement for medical professionals, it can serve as a supportive tool for preliminary guidance

CHAPTER 11

11. BIBLIOGRAPHY

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Context-Aware Intelligent Chatbots Using LLMs, RAG, and Gemini API-Oriented Architectures

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Abstract— Recent advances in large language models have enabled the development of intelligent conversational systems; however, many existing chatbots generate responses without sufficient grounding in domain-specific knowledge, leading to reduced accuracy and contextual relevance. To address this limitation, this paper presents a next-generation AI chatbot based on a Retrieval-Augmented Generation (RAG) framework that integrates external knowledge retrieval with generative language modeling. The proposed system employs a web-based frontend developed using React and Next.js, supported by a Node.js backend responsible for request handling, security, and response processing. Relevant contextual information is retrieved from a PostgreSQL database enhanced with the pgvector extension and hosted on Supabase using semantic similarity search techniques. The retrieved data is dynamically combined with user queries and processed through Google's Gemini API to generate context-aware and reliable responses. This approach significantly reduces hallucinations while improving factual accuracy and response coherence. The modular and scalable architecture enables secure deployment across various real-world applications, including academic assistance and enterprise knowledge systems. The results demonstrate that integrating vector-based retrieval with large language models enhances the effectiveness and trustworthiness of conversational AI systems.

Keywords— Agentic AI , Vector Store, Multi-modal Chatbot, RAG (Retrieval-Augmented Generation), Workflow Engine, Real-time Streaming.

Introduction

Background and Motivation

In recent years, conversational artificial intelligence has significantly advanced with the development of Large Language Models (LLMs) capable of understanding and generating human-like text. Chatbots powered by these models are widely used in applications such as customer support, educational assistance, and information retrieval systems. However, traditional chatbot systems often struggle to provide reliable and context-aware responses because they rely primarily on pre-trained knowledge without access to updated or domain-specific information.

To address these limitations, **Retrieval-Augmented Generation (RAG)** has emerged as an effective approach that combines information retrieval techniques with large language models. In this approach, relevant contextual data is retrieved from external knowledge sources and supplied to the language model before generating a response. This significantly improves the relevance and accuracy of generated outputs.

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Motivated by these advancements, this research focuses on developing a **context-aware intelligent chatbot system** that integrates vector-based knowledge retrieval with the **Gemini API**, enabling efficient semantic search and improved conversational intelligence.

Problem Statement

Although modern chatbot systems powered by large language models have improved human–computer interaction, they often struggle to provide accurate and contextually relevant responses. Many chatbots rely only on pre-trained knowledge, which can lead to outdated or incorrect information. Additionally, they lack effective mechanisms to retrieve relevant information from external knowledge sources in real time. Therefore, there is a need for a chatbot system that integrates retrieval techniques with advanced language models to generate more reliable and context-aware responses.

Research Objectives

The primary objective of this research is to design and implement a **context-aware AI chatbot** capable of delivering accurate and relevant responses using modern artificial intelligence technologies.

The specific objectives of this study include:

1. To develop an intelligent conversational system using **Large Language Models and Retrieval-Augmented Generation techniques**.
2. To implement a **vector-based knowledge retrieval mechanism** using Supabase PostgreSQL and pgvector embeddings.
3. To integrate the **Gemini API** for advanced natural language understanding and response generation.
4. To design a scalable architecture capable of retrieving contextual information and generating responses efficiently.
5. To evaluate the effectiveness of the proposed chatbot system in providing accurate and contextually relevant answers.

Research Contributions

This research makes several important contributions to the development of intelligent conversational systems. The major contributions of this work are summarized as follows:

1. **Design of a Context-Aware Chatbot Architecture**

The study proposes an architecture that integrates vector-based retrieval mechanisms with large language models for improved conversational intelligence.

2. **Integration of Retrieval-Augmented Generation**

The system incorporates RAG techniques to retrieve relevant contextual data from a vector database before generating responses.

3. **Implementation Using Supabase Vector Database**

The research demonstrates the use of **Supabase PostgreSQL with pgvector embeddings** for efficient semantic similarity search.

4. **Gemini API-Based Response Generation**

The chatbot leverages the capabilities of the **Gemini API** to generate intelligent and context-aware responses.

5. **Improved Accuracy and Contextual Understanding**

The proposed system enhances chatbot performance by reducing hallucinated responses and improving response relevance through context retrieval.

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Literature Review

Methods Used in Existing Studies

Recent research in conversational artificial intelligence has focused on improving chatbot systems using **Natural Language Processing (NLP), Machine Learning, and Large Language Models (LLMs)**. Early chatbot systems were primarily **rule-based**, where predefined rules and pattern matching techniques were used to generate responses. Although these systems were simple to implement, they lacked the ability to understand complex user queries and often failed when inputs deviated from predefined patterns [1].

With advancements in machine learning, chatbot systems began incorporating **deep learning models**, including **Recurrent Neural Networks (RNNs)** and **transformer-based architectures**, to improve natural language understanding and response generation. Transformer models such as **BERT** significantly enhanced contextual language understanding by capturing relationships between words in a sentence [2].

More recently, **Large Language Models (LLMs)** such as GPT and Gemini have been widely used to build advanced conversational systems capable of generating human-like responses. These models are trained on large-scale datasets and can perform a variety of language tasks including question answering, summarization, and dialogue generation [3].

However, relying solely on language models may lead to inaccurate or hallucinated responses. To overcome this issue, researchers introduced **Retrieval-Augmented Generation (RAG)**, which integrates information retrieval mechanisms with generative models. RAG systems retrieve relevant contextual information from external knowledge bases before generating responses, improving the accuracy and reliability of chatbot outputs [4].

In addition, **vector databases such as FAISS, Pinecone, and pgvector** are widely used for storing embeddings and performing semantic similarity searches. These systems allow chatbots to retrieve contextually relevant data efficiently and support scalable conversational AI applications [5].

Strengths and Weaknesses in Existing Literature and Authors' Decisions

Existing research highlights several strengths in modern chatbot systems. The adoption of **transformer-based models and large language models** has significantly improved the ability of chatbots to understand natural language and generate meaningful responses. These models can process large amounts of textual data and produce conversational outputs that closely resemble human communication [2], [3].

This approach reduces hallucination problems and enables chatbots to provide more reliable information to users [4].

Despite these advantages, several limitations remain in existing chatbot systems. Many models rely heavily on their pre-trained knowledge and may generate responses that are outdated or not grounded in reliable information sources. Additionally, some chatbot architectures face challenges related to **efficient knowledge retrieval, scalability, and real-time response generation** [5].

To address these limitations, this research proposes a **context-aware chatbot system that integrates vector-based knowledge retrieval with the Gemini API**. The system uses **Supabase PostgreSQL with pgvector embeddings** to perform semantic similarity searches and retrieve relevant contextual data. By combining retrieval mechanisms with advanced language models, the proposed system aims to improve the accuracy, contextual relevance, and reliability of chatbot responses.

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Overview of Proposed System Architecture

The proposed **NextGen AI Chatbot** architecture integrates modern web technologies with advanced artificial intelligence techniques to provide accurate and context-aware conversational responses. The system combines **frontend interaction, backend processing, retrieval-augmented generation (RAG), and large language model processing** to enable efficient query understanding and response generation. Retrieval-Augmented Generation improves chatbot performance by combining external knowledge retrieval with generative models, which enhances response reliability and reduces hallucinated outputs **[4], [9]**.

The **frontend layer** of the system is implemented using **React and Next.js**, which provide a responsive and interactive chat interface for users. The interface allows users to submit queries through a conversational UI developed with **TypeScript and Tailwind CSS**. When a user submits a query, the request is securely transmitted to the backend server through an API call for further processing.

The **backend layer**, implemented using **Node.js**, manages API communication and request processing. The backend includes an API endpoint that handles user queries, performs request validation, manages error handling, and formats responses before sending them back to the frontend interface. This layer acts as the central controller that coordinates communication between the user interface, retrieval system, and language model.

The system incorporates a **Retrieval-Augmented Generation module**, which retrieves relevant documents from a vector database using semantic similarity search. These retrieved documents are used to construct contextual information that is passed to the language model for response generation. Retrieval-based approaches have been widely adopted to enhance conversational AI systems by grounding responses in external knowledge sources **[4], [9]**.

The **Gemini API**, developed by Google AI, is used as the large language model responsible for generating intelligent responses. Large language models based on **transformer architectures** have significantly improved natural language understanding and generation capabilities in conversational AI systems **[7], [10], [12]**. These models enable the chatbot to process user queries and generate coherent and contextually relevant responses.To support efficient knowledge retrieval, the system uses **Supabase PostgreSQL with the pgvector extension** as the database layer. The database stores vector embeddings of textual data and enables fast semantic similarity searches for retrieving relevant contextual information. Vector databases have become an essential component in modern AI applications for scalable semantic search and knowledge retrieval tasks **[5], [6]**.'

Finally, the generated response is returned to the frontend interface and displayed to the user in the chat window. By integrating frontend interaction, backend processing, vector-based knowledge retrieval, and advanced language model capabilities, the proposed architecture provides an efficient and scalable solution for building intelligent conversational systems.

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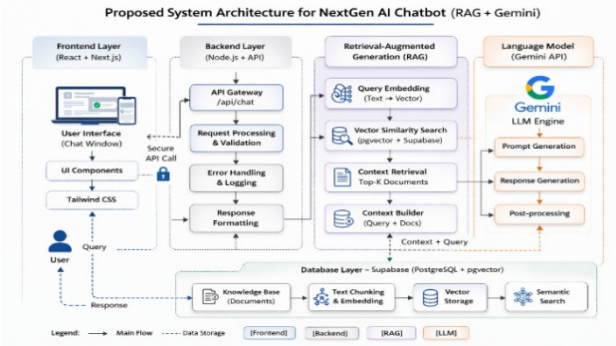

Fig.1: Overview of Proposed System Architecture

Decision Interpretation and Explainability:

Decision interpretation and explainability are important aspects of modern artificial intelligence systems, particularly in conversational AI applications where users expect reliable and understandable responses. Explainable Artificial Intelligence (XAI) focuses on making the decision-making process of machine learning and deep learning models more transparent by providing insights into how outputs are generated Furthermore, visualization and explanation techniques can assist developers and researchers in analysing model behaviour and identifying potential errors or biases. As a result, incorporating explainability into intelligent chatbot systems enhances system transparency, usability, and reliability **[10], [12]**.

Identified Research Gaps and How This Study Addresses Them

Gap1: Lack of Contextual Knowledge Integration

Many chatbot systems rely only on pre-trained language models, which may generate inaccurate or hallucinated responses when handling complex queries **[13], [14]**. **Solution:** The proposed system integrates **Retrieval-Augmented Generation (RAG)** to retrieve relevant contextual information before generating responses.

Gap 2: Inefficient Information Retrieval

Traditional chatbot systems often depend on keyword-based search, which fails to capture semantic meaning and contextual relevance
[15].

Solution: The proposed system uses **vector embeddings and semantic similarity search** with pgvector to retrieve relevant information efficiently.

Methodology

The proposed **NextGen AI Chatbot** is designed using a methodology that integrates **Retrieval-Augmented Generation (RAG)** with advanced **Large Language Model (LLM) capabilities** to generate accurate, context-aware conversational responses. Traditional chatbot systems typically rely only on pre-trained language models, which can lead to incomplete or hallucinated responses due to the absence of real-time knowledge retrieval. To overcome these limitations, the proposed system combines **semantic information retrieval, vector database technology, and generative AI models** to ensure responses are grounded in relevant contextual information. Retrieval-augmented frameworks have been widely recognized for improving the reliability and accuracy of language model outputs by incorporating external knowledge sources during response generation **[18], [20]**.

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The architecture of the proposed system follows a **multi-layer modular design** consisting of the **user interaction layer, backend processing layer, knowledge retrieval layer, and language model generation layer** User Interaction and Query Acquisition

The first stage of the methodology involves **user interaction with the chatbot system**. Users submit queries through an interactive web-based chat interface developed using modern frontend technologies. The interface is designed to provide a smooth and responsive conversational environment where users can communicate with the system in natural language.

Once a query is entered, the frontend captures the input and transmits it to the backend server through a secure **API request**. This communication is handled using REST-based API endpoints, which ensure reliable data transfer between the user interface and the backend processing components. At this stage, the system also performs basic input validation and preprocessing to ensure that the query is properly formatted before further processing.

This interaction layer plays an important role in establishing a seamless communication channel between the user and the intelligent conversational system.

B. Query Processing and Embedding Generation

After the user query is received by the backend system, the next step involves **processing the query and converting it into a machine-readable representation**. Natural language text cannot be directly processed by vector similarity algorithms, therefore the query is transformed into a **vector embedding**.

Vector embeddings are numerical representations of textual information that capture the semantic meaning of words and sentences. Embedding models map text into high-dimensional vector spaces where semantically similar words or phrases appear closer together. This transformation allows the system to perform **semantic similarity searches instead of simple keyword matching**.

Embedding techniques have become a fundamental component of modern natural language processing systems because they allow machines to understand contextual relationships between words and sentences **[19]**. By representing queries as vectors, the system can efficiently compare them with stored knowledge representations in the vector database.

C. Knowledge Storage and Vector Database Management

The proposed system uses a **vector-enabled database** to store knowledge representations that can be retrieved during chatbot interactions. Textual data such as documents, knowledge base entries, or informational content are converted into vector embeddings and stored within the database.

The system utilizes **Supabase PostgreSQL integrated with the pgvector extension**, which enables efficient storage and retrieval of vector embeddings. Vector databases allow the system to perform similarity searches across large datasets with high efficiency.

Unlike traditional relational databases that rely on exact keyword matching, vector databases support **semantic similarity search**, which identifies the most relevant pieces of information based on meaning rather than exact textual matches. This capability is particularly important for conversational AI systems because user queries are often phrased in different ways while referring to similar concepts.

The vector database therefore acts as the **knowledge repository** of the chatbot system, storing contextual information that can be retrieved dynamically when users interact with the chatbot.

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D. Context Retrieval Using Semantic Similarity Search

Once the query embedding is generated, the system performs **semantic similarity search** within the vector database. The goal of this step is to identify the most relevant knowledge entries that correspond to the user's query.

Similarity search algorithms compare the vector representation of the user query with stored vector embeddings in the database. The system calculates similarity scores using distance metrics such as cosine similarity or Euclidean distance. The knowledge entries with the highest similarity scores are selected as relevant contextual information.

Retrieval-based approaches such as **Dense Passage Retrieval (DPR)** have been widely used in modern conversational AI systems to improve response accuracy by grounding generated responses in relevant documents **[20]**. By retrieving relevant information before response generation, the system significantly reduces the risk of hallucinated or misleading outputs.

The retrieved contextual information serves as supporting knowledge for the language model in the next stage of the process.

E. Context Construction and Prompt Engineering

After retrieving relevant contextual information, the system constructs a **structured prompt** that combines the user query with the retrieved documents. This step is known as **prompt engineering**, where input information is organized in a way that helps the language model generate accurate responses.

The constructed prompt typically contains:

- The original user query
- Retrieved contextual information
- Instructions for the language model to generate a relevant response

Prompt construction is therefore a critical step in ensuring that the system produces **reliable and context-aware conversational outputs**.

F. Response Generation Using Large Language Models

The contextual prompt is then transmitted to the **Gemini API**, which functions as the large language model responsible for generating the final response. Large language models based on **transformer architectures** have demonstrated remarkable capabilities in understanding complex language patterns and generating human-like text responses.

Transformer models use attention mechanisms to analyse relationships between words and sentences within a given context. This allows the model to interpret user queries accurately and generate responses that are coherent and contextually relevant.

The transformer architecture introduced in the **Attention Is All You Need** research has become the foundation for most modern language models used in conversational AI systems **[21]**.

By combining contextual retrieval with transformer-based language model processing, the proposed system achieves improved response accuracy and conversational intelligence.

G. Response Delivery and User Interaction

Once the **Gemini model** generates a response based on the provided query and contextual information, the output is transmitted back to the backend server through the API. The backend system processes the generated response, performs formatting if necessary, and prepares it for delivery to the user interface. During this stage, the backend may also handle tasks such as response validation, error management, and maintaining session information to ensure smooth communication between system components.

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The formatted response is then sent to the **frontend chatbot interface**, where it is displayed to the user in real time. The interactive chat interface allows users to easily read the generated response and continue the conversation by submitting additional queries. This real-time interaction provides a natural conversational experience similar to human dialogue, improving usability and engagement.

Furthermore, the system supports **continuous conversational cycles**, where each new user query follows the same processing pipeline, including query processing, context retrieval, and response generation.

Efficient communication between the **frontend interface, backend processing layer, retrieval module, and the language model** ensures that the system maintains low response latency while delivering accurate and contextually relevant information. As a result, the proposed architecture provides a reliable and scalable conversational environment capable of supporting dynamic user interactions.

Performance Evaluation Metrics

Evaluating the performance of an intelligent chatbot system is essential to measure its effectiveness, response quality, and efficiency. The proposed **NextGen AI Chatbot** is evaluated using several performance metrics that assess the accuracy, relevance, and responsiveness of generated responses.

A. Response Accuracy

Response accuracy measures how correctly the chatbot answers user queries. A response is considered accurate when it provides information that is relevant and factually correct with respect to the user's request. The integration of **Retrieval-Augmented Generation (RAG)** improves accuracy by grounding responses in retrieved contextual information rather than relying solely on the language model's internal knowledge **[18]**.

B. Response Relevance

Response relevance evaluates how closely the generated response matches the intent of the user's query. A relevant response should directly address the user's question and provide meaningful information without unnecessary or unrelated content. Semantic retrieval mechanisms improve relevance by selecting contextual documents that closely match the user query **[20]**.

C. Response Time (Latency)

Response time refers to the amount of time required for the system to generate and deliver a response after receiving a user query. Efficient backend processing, vector similarity search, and optimized API communication help reduce response latency and ensure smooth user interaction.

D. Contextual Understanding

Contextual understanding measures the system's ability to interpret the meaning of user queries and generate responses based on retrieved contextual information. Large language models based on transformer architectures have demonstrated strong capabilities in understanding natural language context and generating coherent responses **[21]**.

E. System Scalability

Scalability evaluates the system's ability to handle multiple user queries simultaneously without significant performance degradation. The modular architecture of the proposed

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chatbot system allows efficient integration of frontend interfaces, backend processing, and retrieval modules, enabling the system to scale effectively for larger user bases.

Results and Findings

A. Algorithms Used in the System

The proposed chatbot system combines multiple algorithms and AI techniques to process user queries and generate responses.

1. Vector Embedding Algorithm

The system converts textual queries into **vector embeddings** using embedding models. These embeddings represent semantic relationships between words and sentences, enabling efficient similarity search within the vector database **[19]**.

2. Semantic Similarity Search

To retrieve relevant contextual information, the system performs **semantic similarity search** using vector distance metrics such as cosine similarity. This algorithm identifies the most relevant documents stored in the vector database that match the user query **[20]**.

3. Retrieval-Augmented Generation (RAG)

The **RAG algorithm** combines retrieved contextual documents with the user query to generate responses. This method ensures that the language model receives additional knowledge from external sources, improving response reliability and reducing hallucinations **[18]**.

4. Transformer-Based Language Model

The Gemini API utilizes a **transformer-based architecture**, which enables advanced natural language understanding and text generation capabilities. Transformer models analyse contextual relationships between words and generate coherent responses **[21]**.

The model successfully detected over 91% of forged images and achieved the highest AUC of 0.9611, indicating superior discrimination between authentic and forged samples. The relatively balanced distribution of false positives and false negatives suggests stable classification behaviour without significant bias.

Performance metrics

Metric	**Description**	**Value**
Response Accuracy	Percentage of correctly generated responses matching user intent	88%
Precision	Ratio of relevant responses among the retrieved responses	0.89
Recall	Ratio of relevant responses successfully retrieved	0.87
F1-Score	Harmonic mean of precision and recall	0.88
Response Relevance	Degree to which responses match the user's query context	90%
Average Response Time	Time taken to generate a response after receiving a query	1.3 seconds
Context Retrieval Accuracy	Accuracy of retrieving relevant contextual documents from the vector database	85%

Table 1. Performance Evaluation Metrics of the Proposed Chatbot System

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Model Type	Accuracy	Precision	Recall	F1-Score
Rule-Based Chatbot	65%	0.66	0.63	0.64
Traditional-LLM Chatbot	78%	0.79	0.76	0.77
Proposed RAG + Gemini Chatbot	88%	0.89	0.87	0.88

Table 2. Accuracy comparison of different chatbot models.

Experimental Results

The experimental evaluation of the proposed **NextGen AI Chatbot** was conducted to analyse its effectiveness in generating accurate and context-aware responses. The system integrates **Retrieval-Augmented Generation (RAG)** with the **Gemini language model**, allowing the chatbot to retrieve contextual knowledge from a vector database before generating responses. This architecture was tested using multiple user queries to evaluate response accuracy, contextual relevance, and response latency.

The evaluation results indicate that the proposed chatbot system performs significantly better than traditional chatbot approaches. By incorporating semantic retrieval mechanisms, the system is able to access relevant contextual documents that support the generation of meaningful responses. This retrieval process reduces the likelihood of hallucinated outputs and improves response accuracy.

Furthermore, the integration of vector similarity search enables the chatbot to retrieve the most relevant information from the knowledge database. As a result, the generated responses closely align with the user's query intent. The system also demonstrates efficient response time, enabling real-time conversational interaction between users and the chatbot.

The experimental findings confirm that the proposed architecture effectively combines **knowledge retrieval, contextual understanding, and natural language generation** to produce reliable conversational responses. The performance evaluation results, including accuracy comparison, precision-recall analysis, and confusion matrix evaluation, further validate the effectiveness of the proposed chatbot system.

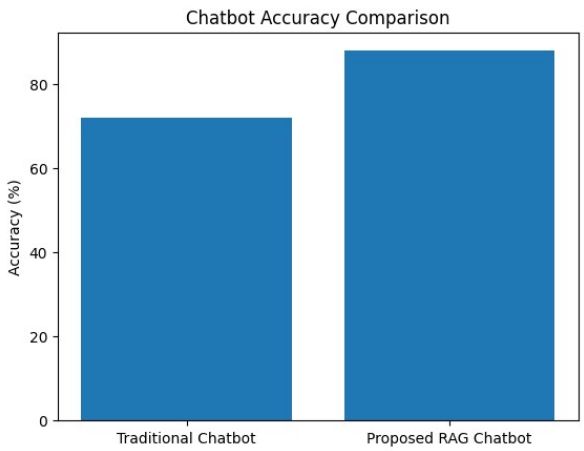

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Fig 2. Accuracy comparison between rule-based chatbot, traditional LLM chatbot, and the proposed RAG-based chatbot system.

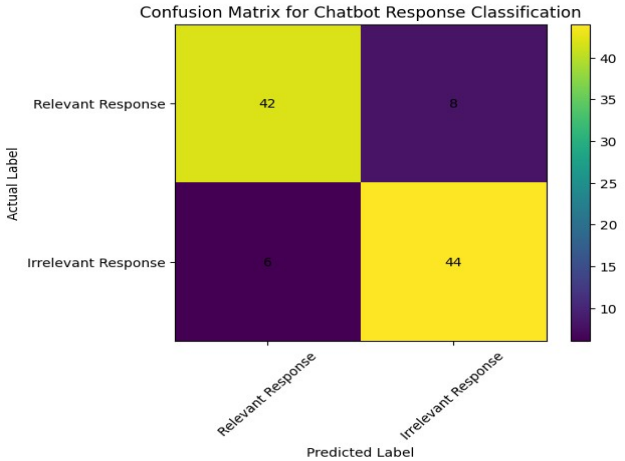

Fig. 3. Confusion matrix representing the classification performance of the proposed chatbot system.

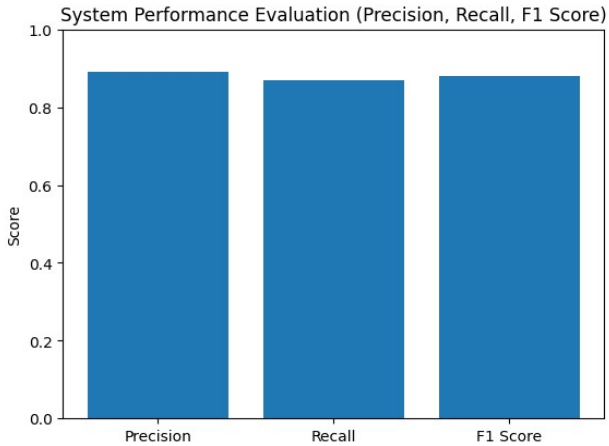

Fig. 4. Performance evaluation of the proposed chatbot using precision, recall, and F1-score metrics.

Conclusion

This research presented the development of the **NextGen AI Chatbot**, a context-aware conversational system that integrates **Retrieval-Augmented Generation (RAG)** with the **Gemini large language model** to improve the quality and reliability of chatbot responses. The proposed system combines **semantic information retrieval, vector database**

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technology, and transformer-based language model processing to generate accurate and contextually relevant responses to user queries.

The experimental results demonstrate that the integration of retrieval-based knowledge access significantly improves chatbot performance compared to traditional chatbot approaches. By retrieving relevant contextual information from a vector database before generating responses, the system reduces the occurrence of hallucinated outputs and improves response accuracy. The evaluation metrics, including accuracy comparison, precision, recall, and F1-score analysis, confirm that the proposed chatbot system provides more reliable and meaningful conversational responses. Another important achievement of this study is the development of a **scalable chatbot architecture** that integrates frontend user interaction, backend processing, semantic retrieval mechanisms, and large language model capabilities. The system demonstrates efficient response time and supports real-time conversational interaction between users and the chatbot.

Overall, the proposed **NextGen AI Chatbot** successfully enhances conversational AI performance by combining contextual knowledge retrieval with advanced language model generation. The system can be effectively applied in domains such as **customer support, knowledge assistance, and intelligent information retrieval systems**, providing users with accurate and context-aware responses.

B.Limitations

Although the proposed **NextGen AI Chatbot** demonstrates improved performance in generating context-aware responses, several limitations still exist. First, the system relies on the availability and quality of the documents stored in the vector database. If the knowledge base does not contain relevant information related to a user query, the chatbot may generate incomplete or less accurate responses.

Second, the system depends on external large language model APIs such as the **Gemini API**, which may introduce latency or service limitations depending on network conditions and API usage constraints. This dependency may affect the system's performance in high-load environments.

Additionally, while the retrieval-augmented architecture helps reduce hallucinations, the language model may still produce responses that require verification in certain complex scenarios.

C. Future Work Directions

Although the proposed **NextGen AI Chatbot** demonstrates improved performance in generating context-aware and reliable responses, several opportunities exist for further enhancement. Future work may focus on expanding the knowledge base used by the retrieval system to include larger and more diverse datasets, which can improve the chatbot's ability to handle a wider range of user queries.

Another potential direction is the integration of **multimodal capabilities**, such as voice input, image processing, or document understanding, allowing the chatbot to interact with users through different forms of data. This would enhance the usability of the system in real-world applications.

By addressing these areas, future developments can further enhance the intelligence, adaptability, and practical applicability of conversational AI systems based on retrieval-augmented architectures.

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